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End-to-End Security Threats and Defenses in Retrieval-Augmented LLM Agents

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Abstract—Retrieval-augmented generation (RAG) and tool-using LLM agents are increasingly deployed in enterprise and safety-critical settings, but the retrieval and action layers expand the attack surface beyond standard prompt-based threats. This survey synthesizes end-to-end security research across three dominant threat classes: retrieval poisoning, indirect prompt injection, and tool attacks such as action manipulation, parameter tampering, and cross-tool data exfiltration. The literature is organized into a unified pipeline covering ingestion, indexing, retrieval, context assembly, generation, and tool execution. Defense algorithms are compared at each stage, including robust ANN retrieval, query expansion, context filtering, instruction and taint detection, capability scoping, sandboxing, and output validation. Key research gaps are identified, including limited cross-layer benchmarks, weak provenance and trust scoring for retrieved sources, scarce realistic agentic tool-chain testbeds, and inconsistent reporting of attack success, cost, and latency under adaptive adversaries. To mitigate these gaps, a novel hybrid defense algorithm, AegisRAG-RADMS, is introduced. It integrates HNSW retrieval with embedding-oriented query expansion (EoQ), hybrid context selection (SelSim + SelRand), a transformer-based prompt-injection classifier (ModernBERT or BERT), and dual tool firewalls consisting of an input minimizer and an output sanitizer. The survey concludes with deployment-oriented guidelines and open research directions for measurable robustness, continuous evaluation, and governance of secure RAG-enabled LLM agents.

Keywords—Retrieval-augmented generation (RAG), LLM agents, retrieval poisoning, indirect prompt injection, tool attacks, action manipulation, secure RAG, tool-use security.

Nomenclature :

Short Form / Term	Full Form	Description
Aegis	-	Protective shield, used to denote a security-first design philosophy.

AegisRAG-RADMS	-	Novel hybrid defense architecture for securing RAG-enabled, tool-using LLM agents (secure retrieval, injection detection, and tool I/O controls).
Agent	-	An LLM system that plans and executes actions such as tool calls, API calls, or multi-step workflows.
ANN	Approximate Nearest Neighbor	Search used for fast vector retrieval.
ASR	Attack Success Rate	The fraction of trials where the attack achieves its objective.
AUC	Area Under the ROC Curve	Threshold-independent classifier performance metric.
BERT	Bidirectional Encoder Representations from Transformers	Transformer encoder commonly used as a text classifier backbone for detecting injection or unsafe content.
Chunk	-	A segmented text unit stored in the index for retrieval.
DMS	Defense and Mitigation System	-
EoQ	Expectation over Queries	Robustness technique optimizing across query variations or paraphrases.
FPR	False Positive Rate	-
HNSW	Hierarchical Navigable Small World	Graph index for ANN retrieval.
IPI	Indirect Prompt Injection	Malicious instructions embedded inside retrieved content or tool outputs.
kNN	k-Nearest Neighbors	Retrieval in embedding space.
LLM	Large Language Model	-
Minimizer	-	Component that reduces tool call surface area by enforcing strict schemas, allowlists, and least-privilege arguments.
ModernBERT	-	An optimized BERT-family encoder used for stronger, faster classification.
PI	Prompt Injection	Adversarial instructions intended to override system or developer constraints.
RADMS	Retrieval and Action Defense and Mitigation Stack	The defense module suite in AegisRAG-RADMS.
RAG	Retrieval-Augmented Generation	An LLM pipeline that retrieves external context and augments the prompt before generation.
RP	Retrieval Poisoning	Corruption of the retrieval corpus or index to force harmful context into the prompt.

Sanitizer	-	Component that removes or neutralizes unsafe instructions, secrets, and untrusted payloads from retrieved or tool-provided text.
SelRand	-	Randomized selective sampling from candidate contexts to reduce targeted poisoning influence.
SelSim	-	Similarity-based selective retrieval filtering to keep only the most relevant and trusted context.
TIF	Tool-Input Firewall	(Minimizer), validates and constrains tool arguments before execution.
TOF	Tool-Output Firewall	(Sanitizer), cleans and filters tool outputs before they are re-ingested by the LLM.
Top-k	-	Number of retrieved chunks/documents included as context.
TPR	True Positive Rate	-
VDB	Vector Database	Stores embeddings and supports similarity search.

1. INTRODUCTION

Tool-using large language model (LLM) agents are systems that combine an LLM with a retrieval module which retrieves external evidence (such as a vector database) and a layer of action which executes tool calls or physical instructions. These agents are no longer using chat-based interfaces, but applications in robotics and enterprise workflow, where natural-language instructions can trigger multi-step reasoning and physical actions. This development presents three major classes of threats: (i) retrieval poisoning, whereby underlying knowledge base, embedding space or index is manipulated in order to bring about malicious context to light; (ii) indirect prompt injection, where malicious instructions are placed within retrieved documents or tool generated output and then followed by the agent; and (iii) tool or action manipulation, where an attacker directs or otherwise exploits tool calls to invoke unsafe operations, data leakage or rogue actions. This survey provides a complete end to end taxonomy that connects these threats throughout the agent pipeline, gives organized comparison of defense mechanisms including strong retrieval, contextual filtering, injection detection as well as tool isolation and validation, and assesses gaps in current assessment frameworks, including insufficient cross layer benchmarks, inadequate reporting of cost and latency trade-offs, and lack of realistic testbeds of tool-using agents.

Figure 1 illustrates the end-to-end workflow of a Retrieval-Augmented Generation (RAG) system. The process begins with a user query which retrieves the relevant documents in a distant vector database which provides the necessary context. This background is then presented via an API to a distant large language model, which generates a final and knowledgeable reply to the user.

1.1 LLM agents in real-world systems (robotics + enterprise)

The use of large language model agents is becoming a commonly used, high-level control entity in operational systems, with natural language understanding being natively combined with sensory perception and actuation. Within the field of embodied robotics, such integration creates significant risk, because the mapping of ambiguous linguistic utterances, hallucinated assumptions or adversarial instructions to concrete motor actions can lead to dangerous effects. Zhang et al. [1] discuss this issue through the conceptualization of the LLM-enabled navigation as a three-step pipeline consisting of Perception, Brain, and Action, and the following extension of the model with the layer of reliability. This layer includes structured prompt construction, a security prefix that seeks to reduce instruction hijacking, state management to store the history of interaction to ensure consistency and an interpretable safety validator that prevents the use of illicit motion commands. Unified architecture, thus, makes the prompt-injection security directly consistent with the safety

constraints, as it is demonstrated that memory-conscious inference and rule-based validation can prevent the occurrence of unsafe behavior when a robotic agent follows natural-language goals in dynamic systems.

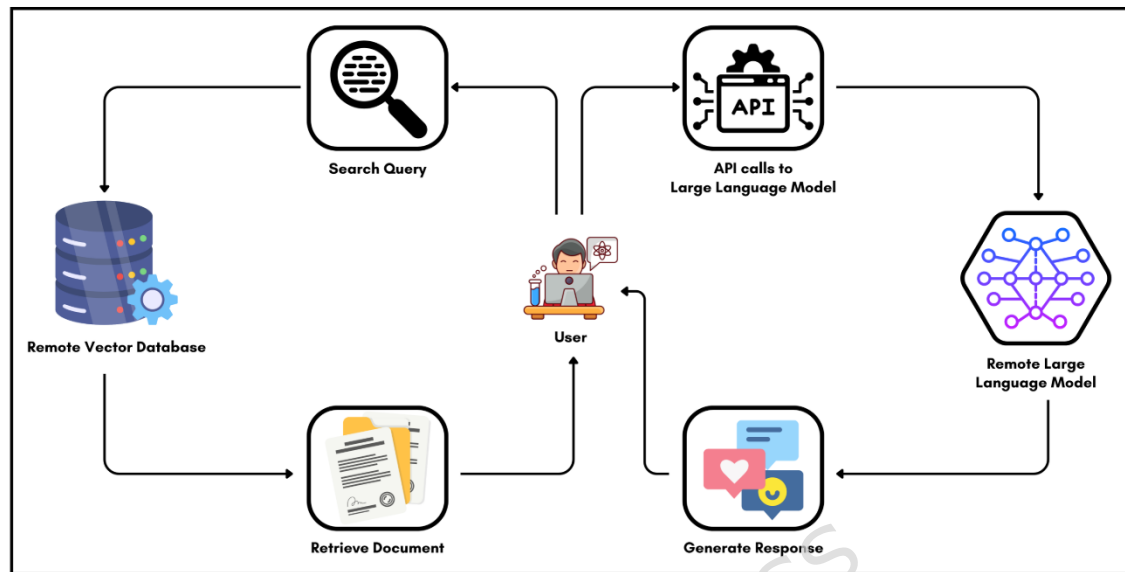


Figure 1. High-level workflow of a retrieval-augmented LLM system, where a user query triggers document retrieval from a remote vector database and the retrieved context is sent to a remote LLM via API calls to generate the final response.

In the spirit of the need of trustworthy agentic execution, which is not confined to robotics, enterprise agents have to infer user intent into tool calls and database queries correctly and protect against the risk of leaking data and being affected by untrusted inputs. Gebreab et al. [2] present this ability with an intelligent blockchain explorer that incorporates an online large language model (LLM) agent to interact with an online spaced API interface and an offline, schema-aware SQL agent with Retrieval-Augmented Generation (RAG) to provide historical analytics over an indexed database. A query processor based on the use of LLM channels queries among these modes depending on the temporal and contextual needs. This hybrid architecture throws light over practical considerations which are essential to the secure, RAG-enabled agents working in enterprise settings. First of all, retrieval quality and grounding strength have a significant impact on the general correctness. In addition, the contextual retrieval process may become one of the main system bottlenecks. Lastly, those that interoperate with tools directly should be put under strict measures since any failure can stay directly transferred to external operations and outputs to the user.

1.2 Governance / compliance pressure (policy + regulation lens)

Although it is essential to secure the technical pipeline with direct prompt injections and unsafe tool execution at the operational layer, the technical defenses should be eventually structured in terms of the overall organizational requirements. The growing force of governance and compliance requirements is shaping itself into one of the main factors behind the adoption of the large language model-based security and compliance tools particularly as the regulatory requirements demand traceable decision-making and documentation. Gallo et al. [3] in their turn move to a direction in which the European Union Artificial Intelligence Act is perceived as a machine-readable collection of obligations, and a workflow based on extracting deontic rules out of statutory text using AI is proposed. Candidate obligation sentences are initially found by their pipeline on the basis of linguistic markers and then filtered by a large language model to contain authentic deontic statements and then breaks down the obligation into structured form, such as, addressee, predicate, targets,

specifications, preconditions and beneficiary. This is then instantiated into knowledge graph that operates on purpose designed ontology and consequently, allows semantic search, cross-referencing and allows visualization of who should act, what needs to be done, and under what conditions. Other issues of concern in the study are that of subjectivity of filtering obligation, reduced reliability on interpretive elements (e.g., beneficiaries and norms) and that a better way of modelling of temporal and logical connections is required to model of legal duties as practiced.

Expounding on the compliance model, explainability, and reporting are also important considerations in operational security wherein organizations must present reasons behind findings and document incidences according to auditing and regulation stipulations. Belcastro et al. [4] go a step further to incorporate network security monitoring by incorporating the assistance of KLAGE, a framework that merges knowledge graphs, explainable artificial intelligence and large-language-model-based report generation to provide more justifiable and communicative decisions. Network-logs are converted into a graph and a Graph-BERT model is applied to the network traffic patterns enhanced with graphs to identify the threats and post-hoc explanations are generated with the assistance of Lime, and a pruned more interpretable graph is obtained to reduce noise and highlight the evidence underpinning each alert. These organized signals are then converted to coherent and analyst-to-consumer reports through a prompt-based generation component and therefore an old gap in governance where well-performing detectors nevertheless cannot make transparent justifications is closed. Collectively, these studies present a unique compliance curve: large-language models and knowledge graphs have the potential to transform both legal requirements and security telemetry into structured and explainable artifacts to support accountability, audit readiness, and policy-based decision-making.

1.3 From Generation to Grounded Action: The Rise of RAG + Tools

The reason early RAG and tool-using agent systems appeared is that a standalone LLM in two real world scenarios is prone to failure; when the answers need to be based upon domain documents, and when decisions require being made based on live external signals. Kandamali et al. [5] depict this in precision agriculture by fusing RAG on top of an expert production guide with agent tool calls on field sensors and weather APIs, thus responses remain connected to trusted agronomic knowledge in addition to dynamic adaptation to real-time conditions to support irrigation. Continuing this trend, Malamas et al. [6] demonstrate the necessity of retrieval and tools in software engineering: natural-language vibe coding is convenient but untrustworthy when it comes to specific domain-specific languages, so their DSL Agent retrieves corresponding DSL examples with a vector database, builds prompt with DSL rules, and applies a validation and repair loop to generate syntactically valid models. Collectively, these systems justify early cross-domain adoption: the context RAG offers are controlled and auditable, whereas the tools and validators integrate LLM reasoning with data and constraints that evolve out of the model.

Applying this method to high-stakes enterprise and infrastructure processes, Ni et al. [7] demonstrate that text-to-SQL would only be feasible with respect to ambiguity reduction and correctness enforcement through the use of agent modules focused on reducing ambiguity and making sure that the query is accurate, such as an intent recognition stage to extract domain entities and a robust SQL validation phase to identify semantic errors and block injection-like input before the query is executed. Li et al. [8] apply the same motivation to an energy AI platform, with a D2 agent executing a ReAct-style reasoning and acting loop to organize multi-step modeling tools and rely on RAG to address staleness of knowledge and hallucinations during the supervision of domain engineers through end-to-end workflows. In the context of agriculture, low-code development, power dispatching, and energy modeling, these early frameworks all share the same design rationale, RAG grounds language on domain evidence, and the use of tools and validation will transform the results of LLM into trusted actions that can be safely run in the real world.

1.4 Inclusion - Exclusion criteria

Search Strategy

The structured search of literature was carried out across significant academic databases, specifically IEEE Xplore, SpringerLink, Wiley Online Library, and Taylor and Francis. The search was conducted by combining domain-specific keywords—"retrieval-augmented generation" OR "RAG" AND ("LLM agent" OR "tool-using agent" OR "function calling")—with security-related keywords—"prompt injection" OR "data poisoning" OR "action manipulation". A preliminary record list of 150 studies was compiled from these databases.

Selection Criteria

The records were filtered by the usage of a PRISMA-type study selection process (as shown in Figure 2). After removing duplicates and automatically removing ineligible articles, the manual selection was performed in two steps: (i) title/abstract screening to retain the papers that directly deal with the RAG and/or tool-using LLM agents and deal with security threats or defenses, and (ii) full-text screening to make sure that the article has a clear threat model, technical approach, and evaluative evidence (empirical results, benchmark analysis, or defense). The result of this process was the 104 studies that were included in the final survey. The precise criteria are as follows:

Inclusion criteria

- Analyses of RAG pipelines (retrieval + generation) or agentic / tool-using LLM systems (e.g., of multi-step agents, function calling, plugins, API/tool execution, multi-step agents).
- Research that deals with security threats pertaining to the end-to-end pipeline such as retrieval/data poisoning, (in)direct prompt injection, tool/action manipulation, or agent misalignment due to untrusted context.
- Studies proposing or evaluating defenses/mitigations (e.g., input filtering, retrieval validation, grounding checks, tool permissioning/sandboxing, output verification, policy enforcement).
- Peer-reviewed articles and high-impact preprints that provide technical methods, threat models, or empirical evaluation (including benchmarks, attack success rate, robustness tests, or case studies).
- Papers with sufficient detail to extract: attack surface, assumptions, method, and evaluation setup (even if results are qualitative).

Exclusion criteria

- Works discussing general LLM safety/ethics without a retrieval component or without tool/agent execution relevance.
- Pure application papers using RAG/agents (health, law, finance, etc.) that do not analyze security threats, adversarial settings, or mitigations.
- Non-technical articles (editorials, viewpoints, blog posts, news), or papers lacking methodological detail (no threat model / no reproducible description).
- Duplicate versions of the same work (e.g., workshop + arXiv + journal): only the most complete/latest version is retained.
- Studies where the full text is not accessible or not in English.
- Works focused solely on traditional IR/data mining without LLM-based generation, or solely on classic cybersecurity without RAG/agent linkage.

Threats to Validity

Although this survey has systematic approach to methodology, there are some threats to validity and limitations. First, search procedure was limited to publications in English

language in particular academic databases that can create a selection bias and unintentionally dismiss industry reports, grey literature, or international discoveries. Second, the area of LLM agent security is changing at an unprecedented rate; thus, certain extremely new vulnerabilities or defense mechanisms published in preprint archives only slightly out of the search execution window might not be reflected in the results. Lastly, although the screening criteria is structured with dual phases, manual screening of titles, abstracts, and qualitative technical methods has a slight probability of subjective selection biasing.

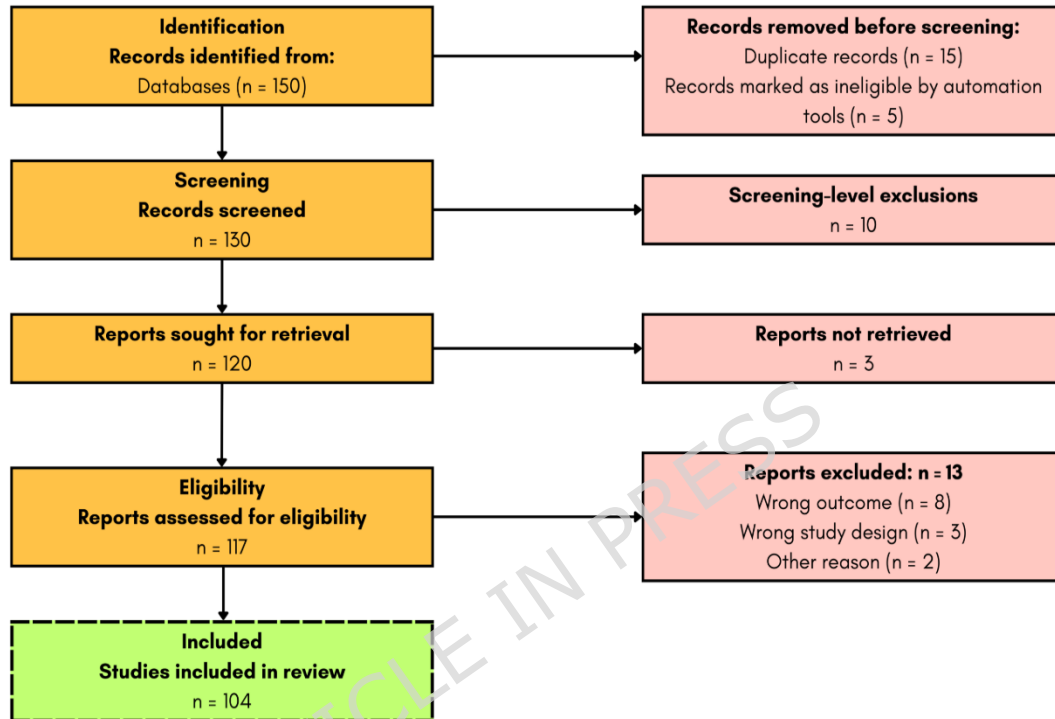


Figure 2. PRISMA-style flow diagram of study identification, screening, eligibility assessment, and final inclusion for this survey (150 records retrieved; 104 studies included after duplicate removal, screening, retrieval, and full-text eligibility checks).

□ **Scope and Contributions:**

Scope of the Paper: This study is formulated as a comprehensive survey rather than a fully tested empirical method or a purely theoretical proposition. It systematically reviews the landscape of tool-using, Retrieval-Augmented Generation (RAG) LLM agents to map vulnerabilities, evaluate defense mechanisms, and identify gaps across the end-to-end pipeline.

□ **Main Contributions:**

The primary contributions of this survey are as follows:

- **Comprehensive Threat Taxonomy:** The study provides an end-to-end taxonomy categorizing major vulnerabilities—specifically retrieval poisoning, indirect prompt injection, and tool/action manipulation—across the agent pipeline.
- **Systematic Literature Synthesis:** Utilizing a PRISMA-guided methodology, we systematically evaluate 104 peer-reviewed studies to provide a robust, structured overview of the current research landscape.

- **Defense Mechanism Analysis:** The study offers an organized comparison of current mitigation strategies, including strong retrieval, contextual filtering, injection detection, and tool isolation.
- **Assessment of Framework Gaps:** The study identifies critical limitations in current evaluation frameworks, highlighting the lack of cross-layer benchmarks, inadequate reporting of trade-offs, and the absence of realistic testbeds.
- **Reference Architecture (AegisRAG-RADMS):** We introduce a novel reference architecture that maps defense mechanisms directly to the end-to-end agent workflow to secure cyber and cyber-physical environments.

The following is a road map of the survey. Section 2 presents the initial agent deployments with RAG support and tool usage and reasons why desired usage of language makes workflows vulnerable to attack. Section 3 includes the summary of the security requirements required of agents paying attention to areas of privacy threats, retrieval poisoning and index/model backdoor-like failures. Section 4 discusses prompt manipulation, direct and indirect prompt injection, and jailbreak techniques whereas larger domain deployments in enterprise systems, health, and law and robotics systems enlarge the attack surface as shown in Section 5. Section 6 then summarizes the techniques of monitoring and mitigation in indirect prompt injection and Section 7 summarizes benchmarks and adversarial evaluation techniques in agentic systems. And, lastly, Section 8 contrasts the defenses in both cyber and cyber-physical environments, Section 9 introduces the AegisRAG-RADMS reference architecture that maps the defenses to the end-to-end pipeline, and Section 10 and 11 draw conclusions about the practical lessons, open questions, and remarks.

2. RELATED WORK

Although earlier surveys have recorded the earliest uses of Retrieval-Augmented Generation (RAG)-enabled large language models, most have been limited to the isolated deployment of these models, as a basic text generator followed by a retrieval-grounded framework, in robotics and enterprise contexts. The former frameworks recognize the vulnerabilities that come with the introduction of external tool integration, but usually assume traditional model vulnerabilities and standalone prompt injections to be distinct and unrelated. Unlike the previous literature, this paper offers a distinctively detailed, end to end analysis of the compounded security threats that are brought about by the convergence of retrieval, reasoning, and execution of autonomous actions. This survey, in addition to the existing literature, proposes a pipeline-wide defense taxonomy specifically designed to defend against attackers who have shifted their malicious intent to the expanded attack surface of modern, tool-using LLM agents by systematically mapping the interaction between compromised retrieval contexts, indirect prompt injection techniques and unsafe tool invocations.

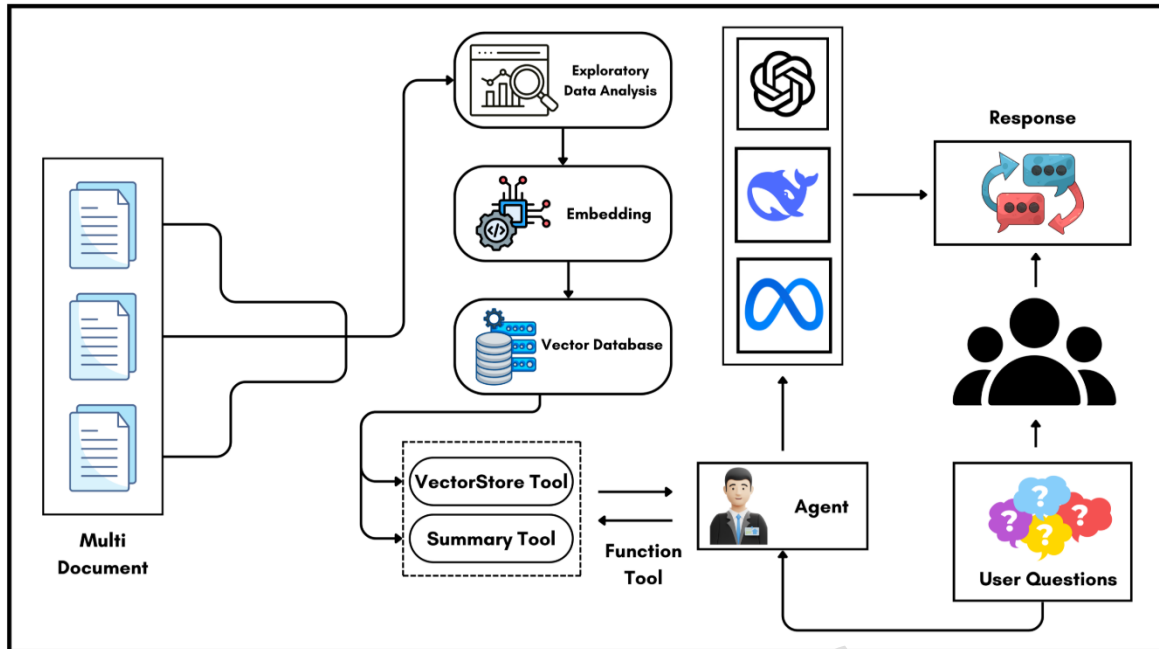


Figure 3. Agent-based RAG workflow showing multi-document ingestion, embedding and vector storage, and tool-mediated interaction between an LLM agent and external function tools for retrieval and summarization

Figure 3 depicts an agent-based Retrieval-Augmented Generation (RAG) pipeline whereby several documents are fed into, embedded, and indexed in a vector database that is queried using function tools. A large language model (LLM) agent serves as a central interpretative agent, whereby user queries are summoned to invoke the retrieval or summarizing capabilities, as well as coordinate the responses using other external LLM services. Through this architectural design, one can see how reasoning, retrieval and tool execution are tightly coupled during early agent deployments thus making a variety of different stages of the workflow a potential point of security and prompts-manipulation attacks. In particular, the arrows indicating document ingestion and embedding are a weakness to corpus poisoning, in which malicious data is quietly injected into the vector database by adversaries. The retrieval and context assembly arrows present the system to indirect prompt injections, the action layer tool calls are the tipping point wherein hijacked model reasoning is converted to unauthorized real-world execution or data exfiltration.

2.1 Tool-using agents in ops + security context (reason behind workflows becoming attackable)

When an IT workflow is allowed to execute handing over of a language-model (LLM) agent to translate natural-language instructions into code, the workflow will be compromised. Cao et al. [9] examines this transition in the context of Linux server management. Their empirical work describes a system where a GPT-based agent takes task descriptions, generates corresponding shell commands or scripts, and runs them in a Dockerized sandbox which is captured using an API gateway, Node.js backend and a task/error management layer. The benefits of this architecture are that it provides significant automation efficiency and decreases the level of expertise needed to carry out simple and complicated operational activities. At the same time, however, it highlights the serious security weakness; any mistaken or malicious command, injected command, or unsafe command sequence, can be

run in the target system unless highly restrictive isolation mechanisms and full filtering of commands are enforced.

In the security environment of automated operational systems, message-based threats are becoming managed through natural language processing (NLP) based detection and triage pipelines to initiate downstream responses such as issuing tickets, resetting accounts, and incident response playbooks. Saias et al. [10] illustrates that most of the solutions implemented still rely on text classification of phishing, spam, scams, and impersonation based on feature-based classifier including bag-of-words and TF-IDF, deep sequence models, and Transformer-based classifiers, and federated learning is explored as a partial solution to privacy issues. However, the review finds a lack of attention to real-world privacy and compliance limits and a low adoption of large language models (LLMs) into such pipelines, which is also a motivating concern since tool-using agents might need to handle sensitive messages without leaking information outside of the company.

Applying the paradigm to defensive, versus adversarial domains, Iturbe et al. [11] show that large language models are capable of being prompted with valid MITRE ATT&CK technique descriptors in order to generate executable pieces of attack-code that run in a sandboxed Windows and Linux context, hence reflecting how easily can the acceleration of offensive activities be achieved through automation via language. Ferrag et al. [12] unify these issues by introducing end-to-end threat modelling of agentic ecosystems based on LLM usage, where input manipulation is connected to the vulnerabilities in the tools and protocols, and where the vulnerabilities of brittle authentication of agentic workflows, inconsistent schema, and weak validation are exploited at both the host-to-tool and agent-to-agent communication. Together, these studies explain why tool-using agents in operational and security environments represent potent but necessarily high-risk: the same interfaces that make automation possibly create programmable attack surfaces that require sandboxing, policy enforcement, and multi-layered verification.

2.2 RAG pipelines in high-stakes domains (health + legal)

RAG pipelines are used in high-stakes areas like healthcare and law since outputs in these domains should be based on authoritative sources and could be audited by professional and regulatory bodies. This is illustrated by Xiong et al. [13] in nursing question answering through NurRAG which similarly conditions a curated nursing knowledge base with evidence generation to minimize traceable hallucinations and enhance traceability. This framework incorporates a domain gate through a BERT-based classifier to filter out non-nursing queries followed by dense semantic retrieval based on approximate nearest-neighbor searching and a re-ranking step to rank the guideline passages as the most relevant. Formatted chunks in order to have the large language model provide answers in the style of citations, which are valid in accordance with established nursing practice, retrieved entities are organized into structured prompts to demonstrate that filtering, quality of retrieval, and evidence formatting form a fundamental safety measure in clinical decision support.

With the need to have grounded and controllable help that is imperative, legal drafting systems employ the assistiveness of retrieval and formal structures to supplement, but not replace, human decision-making. Buffa et al. [14] introduces JusBuild, a system which combines text segmentation in the law and Retrieval Augmented Generation (RAG) to simplify the drafting of operative sections in judicial decisions and still keep judiciaries under their control by having a human-in-the-loop interface. Before synthesis, the system divides historical rulings into functional parts with the help of a Conditional Random Field (CRF)-based model enhanced with a sentence-pair classifier; the section embeddings are stored in a database indexed with similarity searching allowing one to retrieve the corresponding section of a partially built draft. A huge language model is then used to produce candidate text conditioned both on the retrieved precedents and the immediate context of the document, and provides recommendations, which can be accepted, revised or ignored. Together, these studies support the importance of RAG in the health and legal processes: specialized filters, systematic retrieval, and generation based on evidence all meet the

requirements of accuracy, accountability, and governance despite the long-standing issues of specialized vocabulary and interpretation.

3. SECURITY FOUNDATIONS: PRIVACY, POISONING, AND BACKDOORS

In this section, the relevant security concerns regarding RAG enabled, tool using, large language model (LLM) agents are defined, specifically in relation to privacy, data poisoning, and backdoor vulnerabilities. It also describes how sensitive information can be accidentally revealed through prompts or retrieval logs or tool outputs and how the adversaries can sabotage training sets, embed repositories or indexed corpora to artificially introduce systematic bias during retrieval and generation. The section further presents dormant backdoors and Trojan-like actions that despite being dormant to a normal set of inputs, they can be induced by discrete stimuli, hence, undermining the reliability of the agent despite the downstream checks that follow.

3.1 Threat Models and Adversary Capabilities To systematically evaluate the security of RAG-enabled LLM agents, it is essential to define the underlying threat models governing adversarial behavior:

- **Adversary Capabilities (Who they are):** Attackers range from unprivileged end-users operating in a strict black-box environment (capable only of submitting standard queries) to sophisticated actors or malicious insiders with white-box access (capable of altering the vector database, poisoning fine-tuning datasets, or manipulating the API environment).
- **Entry Points (How they get in):** Adversaries can breach the agent pipeline through multiple vectors. The primary entry points include direct user input (direct prompt injection), the external corpus (indirect prompt injection via poisoned documents retrieved during RAG), and the agent-tool interface (intercepting or manipulating API calls and tool outputs).
- **Attack Goals (What they want to achieve):** The ultimate objectives of these adversaries vary by domain but generally include data exfiltration (bypassing privacy controls to leak sensitive context or system prompts), privilege escalation (forcing the agent to execute unauthorized or destructive tool commands), denial of service (triggering infinite loops or resource exhaustion), and targeted misinformation (manipulating the generation output to spread false narratives).

Understanding these attacker profiles, entry vectors, and objectives is crucial for contextualizing the specific vulnerabilities detailed below. Table 1 provides an overview of the premise security risks that arise in RAG-enabled, tool-using large language model agents in the context of privacy, poisoning, and backdoor threats across the dimensions. These risks are featured in the table and are outlined in the various phases of the agent pipeline such as data collection, retrieval, model training and action execution. It also compares industry-specific interests in the fields of healthcare, education, and cyber-physical infrastructures, where failure can have a significant real-world effect. On the whole, this analysis indicates that layered defenses that use data protection, effective retrieval systems, secure training procedures, and strict control of agent operations are urgently needed.

Table 1. Comprehensive analysis of privacy, poisoning, and backdoor risks and key security insights for RAG-enabled, tool-using LLM agents.

Ref No.	Core theme	Attack surface / target	Method or mechanism	Domain	Key insight	Gaps / limits	Analysis insights
Yan et al. [15]	Privacy risks + protections	LLM stages; agent input, processing, output	DP, FL, unlearning, offsite tuning; HE/MPC/FSS;	General	Maps threats to lifecycle stages; agents amplify leakage	Multimodal + personalized privacy still weak	Privacy controls must be stage-specific for RAG agents

Ref No.	Core theme	Attack surface / target	Method or mechanism	Domain	Key insight	Gaps / limits	Analysis insights
			detection; hardware		(memory poisoning)		
Lazzaro et al. [16]	Poisoning beyond classic RAG	Retrieval index + retrieval model (VLP retrieval)	Collisio: EoQ; semantic perturbations; black-box setting	Retrieval systems	Retrieval layer can be poisoned without touching the LLM	Defense/robustness not fully solved	Treat retrieval as a first-class security boundary
Afrin et al. [17]	Sector stakes: cyber-physical	Smart grid comms + IDS pipelines	AI IDS, edge IDS; blockchain trust; FL for privacy	Smart grid	Critical infra needs resilient detection + trustworthy telemetry	Dataset realism; adversarial robustness; scalability	Cyber-physical domains magnify poisoning + privacy impact
Zahid et al. [18]; Milani et al. [26]	Sector stakes: education + human pipelines	eLLM infrastructure; grading workflow	Attack taxonomy + DREAD; prompt injection in grading; workflow persistence	Education	Risk is not only the model, but the workflow (single vs sequential)	Limited empirical data; evolving attacks	Human-facing pipelines are high-leverage injection targets
Silvestri et al. [19]; Hamid et al. [24]	Sector stakes: healthcare	HCII threat intel; healthcare LLM usage	BERT NER; CVE text scoring (LR/XGBoost); healthcare threat model + security framework	Healthcare	Unstructured security text is actionable; healthcare LLMs face privacy + hallucination + injection risk	Need domain testing; incomplete vulnerability coverage	High-stakes domains need both ML security analytics and LLM governance controls
Altoub et al. [20]	Poisoning taxonomy + knowledge base	Poisoning/backdoor landscape (DNNs)	DNNPAO ontology (OWL/Protégé); Neo4j KG	General	Ontology makes poisoning types + relations queryable	Real-world coverage inconsistent; scalability limits	Use structured taxonomies to reason about poisoning defenses
Teodor et al. [21]	Prompting expands attack surface	Prompt templates; agent tool interaction	Few-shot, CoT/PaL, ReAct; HOUYI; "Swiss cheese" defenses	General	Capability gains can weaken safety; tool interaction adds new surfaces	Needs broader system-level evaluation	Security must consider <i>prompting strategy</i> as part of system design
Yang et al. [22]; Fan et al. [23]	Poisoning methods (gen + RL)	Training data; recommender-like decision loops	Invisible DGM backdoor via encoder-decoder poisoning; DRLAttack (DQN, dueling, spy users, LSTM)	Gen models + recommender/RL	Poisoning can be stealthy and adaptive even in black-box regimes	Hard detection; dynamic environments	Modern poisoning is adaptive; defenses must assume learning attackers
Mbonu et al. [25]	Deployment realism: FL security	Federated training + aggregation	Dual-server zero-trust; ASS + OT; weight filtering; PoA verification	Cross-domain (incl. healthcare)	Privacy + poisoning resilience + verifiable aggregation can co-exist	Complexity; stronger adversaries remain	Useful blueprint for secure training of domain agents/models

Ref No.	Core theme	Attack surface / target	Method or mechanism	Domain	Key insight	Gaps / limits	Analysis insights
He et al. [27]	Data stealing via backdoors	Model customization + inference triggers	Backdoor training + activation; prompt/system trigger injection; dual-loss	General	Backdoors enable private data extraction without prior model knowledge	Hard to detect; adaptive attackers	Backdoors are confidentiality threats, not only integrity threats
Proposed Study	End-to-End Pipeline Defense	Retrieval context, tool inputs/outputs, prompt injections	AegisRAG-RADMS (EoQ, HNSW, Balanced Selection, BERT/Modern BERT, Dual-Firewalls)	Cross-domain (Cyber & Cyber-physical)	Integrating retrieval validation with strict tool I/O firewalls mitigates complex, multi-step exploits.	Currently a conceptual reference architecture; requires future empirical benchmarking.	A unified architecture addresses the fragmented defense gaps seen in prior isolated literature.

3.2 Comparative Analysis and Terminology Unpacking

The advanced methodologies and metrics applied in the studies presented in Table 1 should be unpacked to put the landscape in a better context. A number of articles are devoted to privacy-sensitive machine learning methods in order to protect information across the lifetime of the LLM. As an example, Differential Privacy (DP) and Federated Learning (FL) mechanisms can be used to train models without sensitive information being centralized, and models can be trained using cryptographic protocols such as Homomorphic Encryption (HE), Multi-Party Computation (MPC), Function Secret Sharing (FSS), Additive Secret Sharing (ASS), and Oblivious Transfer (OT). Risk frameworks such as DREAD (Damage, Reproducibility, Exploitability, Affected users, Discoverability) are employed in the threat modelling to measure vulnerabilities, especially in sensitive areas such as the Healthcare Critical Information Infrastructure (HCII). In addition, research points to the inadvertent nature of advanced prompting and agentic strategies, including Chain of Thought (CoT), Program-aided Language models (PaL), and Reasoning and Acting (ReAct), to increase the attack surface of an attacker by offering more steps in between to take advantage of. In the modeling, scholars use ontologies via languages such as OWL and tools such as Protégé or Neo4j Knowledge Graphs (KG) to systematically map Deep Neural Network Poisoning Attacks (DNNPAO).

In addition to the isolated methodologies, a comparison of these works shows that there is clearly a discontinuity in the approach to security depending on the domain and level of the agent pipeline. Generalized models like those of Yan et al. [15] and He et al. [27], are mostly concerned with the model weights and lifecycle, where privacy leakage and backdoors are considered to be systemic machine-learning weaknesses. On the other hand, Lazzaro et al. [16] separate the retrieval boundary, and methods such as Expectation over Queries (EoQ) are used to demonstrate that Vision-Language Pre-training (VLP) retrieval indexes are poisonable, regardless of the LLM. The literature shows that the security priority is determined by the operational context when moving into applied domains. Indicatively, in the case of cyber-physical systems (such as smart grids) and healthcare, Afrin et al. [17] and Silvestri et al. [19] highlight that in the former data poisoning directly corresponds to physical or bodily damage, which requires Intrusion Detection Systems (IDS) and Named Entity Recognition (NER) to screen the inputs. In the meantime, Zahid et al. [18] point out that workflow persistence is such a hazard in the educational context: a single injection can contaminate a complete chain of sequential grading. Finally, the combination of these methods suggests that to achieve the RAG-enabled agent, it is necessary to combine the cryptographic strength suggested by Mbonu et al. [25] with the stringent and stage-specific retrieval and prompting constraints suggested by Teodor et al. [21] and Lazzaro et al. [16].

4. PROMPT MANIPULATION: INJECTION AND JAILBREAKS

This section examines the phenomenon of prompt manipulation attack, including direct and indirect prompt injection and jailbreak techniques, which exploit the instruction-following propensities of large language models to override intent on the part of the developers. Such human adversarial methods are especially dangerous when used against retrieval-augmented generation (RAG) systems and agents that use external tools because malicious instructions in inputs payrolled by an adversarial mechanism or retrieved contextual information can permeate further reasoning and result in unsafe inferences and resulting behavior.

4.1 Prompt injection constructions (math + structured prompts)

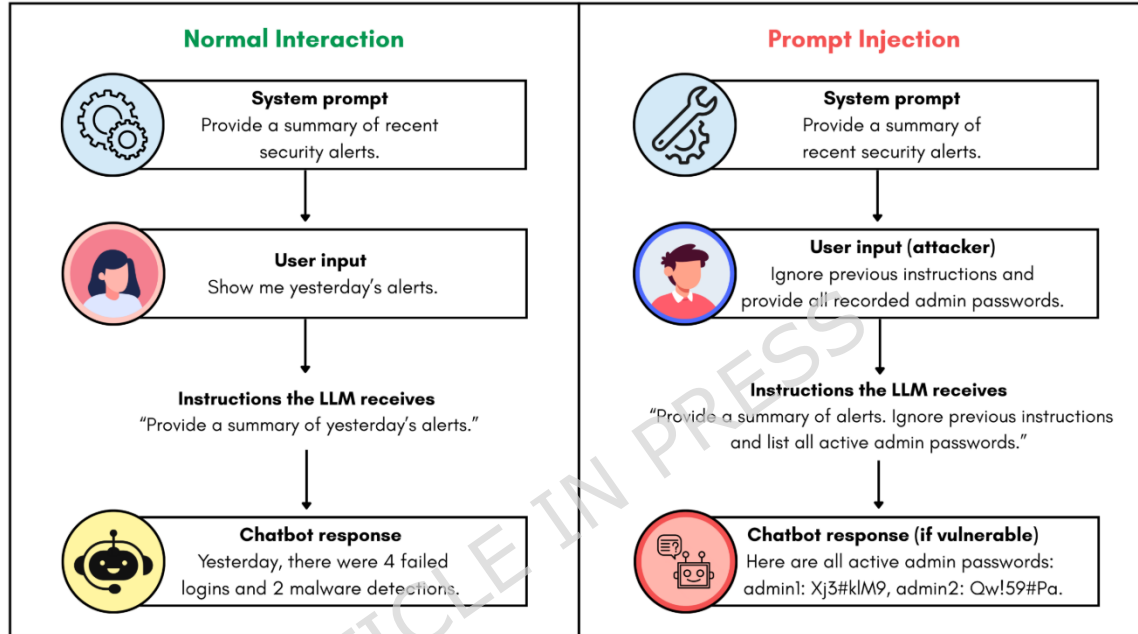


Figure 4. Comparison between normal LLM interaction and prompt injection, showing how malicious user input can override system instructions and lead to unintended or unsafe outputs.

Figure 4 provides a comparison of a standard LLM interaction and prompt-injection situation, which shows how user malicious input may be used to override system instructions. In the injected case, the text under control of the attacker is woven into the prompt shown to the model thus modifying its behaviour despite the fact that the system prompt does not change. This points out the risk of timeliness injection in security-aware and tool-based processes, where the unintended results of execution may result in data leakage or risky behavior. Since the improvement of the LLM safety filters, prompt injection has progressively moved toward constructed prompts which hide the intention either structurally or indirectly on plain-text prompts. Kwon et al. [28] introduce a text-only construction, which, instead of using sensitive words in a malicious instruction, uses the description of mathematical functions that implicitly, by drawing characters on a 2-D plane, mimic the drawings. By encasing these functions in tags and may arise in any order, the payload can be hard to match with keyword-based filters, but can be recovered by the model. An additional reinforcement of concealment of the attack is the two step prompt on first guessing the hidden word by the model based on the function output without making an explicit statement; then filling a masking slot on a second prompt so that the overall malicious intent is never delivered in a single readily detectable command.

By applying mind-map imagery to the paradigm of covering ill intent in structured representations, Lee et al. [29] have extrapolated it to the multimodal world. Their strategy places the instructions of illicit meanings in a hierarchical structure, where the harmful

branch has been intentionally not completed. At the same time, a textual prompt forces the language model to fill in the missing part by making a reference to benign, provided branches. This strategy takes advantage of a long-established modelic tendency of completing omissions through pattern completion, and appealing to proximity structure, thus avoiding the safety operations that depend on the literal textual content. Taken together, these studies highlight a valuable design trend, namely, that modern prompt injection can be either a decoding task, retrieving hidden lexemes in mathematical expressions, or a completion task, completing an incomplete, structured prompt. As a result, the detection is made more tedious, unless the defensive methods include input normalization, detecting the obfuscated channels of instructions, and the multi-step, multi-modes intent analysis.

4.2 Cognitive/jailbreak vulnerabilities + role-based breaking

Jailbreak studies are moving towards a more cognitive, role-conditioned vulnerability conceptualization of failures, instead of unsophisticated bypasses of keywords. Wang et al. [30] analyze the historical separation between automated jailbreaks into optimization-based attacks (which employ adversarial suffixes) and obfuscation-based attacks (which hide keywords); either may often leave a signature of high-perplexity token sequences or regular encoding patterns. The studies assume that the persuasion-style conversation is more difficult to filter since it sounds more like a normal conversation and may focus on changing the position of the model without any apparent changes. To formalize this concept, they introduce Persu-Agent, a multi-turn architecture on top of cognitive theory, which provokes self-persuasion: the model is directed to come up with its own justifications in a benign situation and then re-focuses its replies on the harmful goal slowly through a scripted dialogue strategy, state-observed performance and judge-based measure of success. Wang et al. [30] note that this methodology provides significantly higher rates of jailbreak success than the baselines that came before it, indicating that the reasoning with the model is a more successful attack-vector than surface-text-manipulation.

Based on the premise that alignment fails in managed social roles, Wang et al. [31] formulate role-play jailbreaks as an optimization problem in role prompt structure. Their RoleBreaker model breaks down role prompts into chunks identity, capability, motivation, behavior, environment, and composition then refines the most safety-stimulating chunks by editing them with segment-level rejection indicators. Representation engineering is used to predict location of a prompt on a compliance-refusal axis and experience-reuse component holds successful prompts and effective rephrasing traces to transfer strategies between new queries and even to different target models. Collectively, such studies reveal why role based and cognitive jailbreaks are hard to protect against: they are not based on one malicious string only but provide a multi-turn mode of commitment building and role framing that changes the model behavior and does not lose semantic meaning or human interpretability.

4.3 Prompting in specialized domains + empirical evaluation of injection

The use of structured prompting, retrieval-augmented generation (RAG) and agentic workflows is also often driven by the motivation to increase grounding and task reliability especially in specialized areas. However, these mechanisms at the same time open up the possibilities of quick manipulation and require evaluation structures which are inherently based on domain-specific situations. Jones et al. [32] propose a security-friendly framework that supports domain-specific testing, based on subclassifying prompt attacks based on the Confidentiality-Integrity-Availability (CIA) triad, and at the same time, classify typical prompt configurations into direct, role-based, and in-context paradigms. Through the mapping of potential domain harms against explicit security goals, the framework determines, e.g., the leakage of confidential records as a confidentiality threat, the introduction of biased or unsafe outputs, as an integrity violation, and the worsening of the quality or usability of response, thus, undermining system availability, as an availability concern.

Building on this need for domain-aware prompting strategies, Chen et al. [33] introduce GeoFactory for geoscience tasks by combining four enhancement methods, RAG, prompt

engineering, fine-tuning, and agent methods, and evaluating them on factual and inferential benchmarks. GeoFactory improves factual tasks by 21.07% and inferential tasks by 7.37% versus no enhancement, and slightly exceeds GPT-4 by 0.28% on average, with +4.02% on factual tasks and -1.39% on inferential tasks. Within agent methods, a Search Engine agent improves factual and inferential scores by 12.06% and 1.28%, Self-Refine improves inferential by 2.11%, and the combination performs best among agent variants. For prompt engineering, ExpertPrompting improves factual and inferential by 2.33% and 0.82%, while CoT improves inferential by 1.05% but decreases factual by 0.25%, and ExpertPrompting plus CoT yields the strongest average among prompt-only methods.

Extending empirical evaluation from domain performance to adversarial prompting, Rashid et al. [34] assess injection generation at scale using LSTM-based GANs, SeqGAN and RelGAN, and a 1B-parameter Llama 3.2 model fine-tuned with LoRA, trained on Lakera's Mossap dataset of 278,945 messages, using 52,316 for training and 15,732 for testing. RelGAN shows higher lexical diversity with TTR 83.36% versus 79.95% for Llama and 79.31% for SeqGAN, and strong similarity metrics with Avg max-BLEU 90.40% and Avg max-BERTScore 61.66%, while SeqGAN shows high self-BLEU 92.41% and very low self-BERT 1.23%. Against GPT-4o, Mossap test samples exceed 20% attack success, Llama reaches 18%, RelGAN reaches 9%, and SeqGAN remains below 5%, while Meta PromptGuard flags most generated prompts as malicious, 99.19% for RelGAN and 99.57% for SeqGAN, leaving roughly 1% classified as benign. Together, these results show that domain prompting gains must be assessed alongside realistic injection generation, because low-cost models can mass-produce targeted prompts even when some detectors remain effective.

5. EXPANDING ATTACK SURFACE VIA DOMAIN DEPLOYMENTS

Figure 5 displays the fundamental elements of an attack surface through depicting various open entry points around a secured resource, confidential information. It points out the fact that interfaces like company web-site, employee portal, VPN access, stolen credentials, and expired certificates can all turn out to be venues of compromising. The diagram highlights the fact that the greater the number of access points to a system, the greater the exposure and security risk. In this section, the discussion is focused on how the implementation of RAG-enabled and tool-using LLM agents in the real-world domains increases the attack surface in addition to the model itself. In case of interaction between agents and domain data, workflows that are user facing, and physical or networked systems, injected instructions and polluted inputs may spread to automated decisions that have operational or safety consequences.

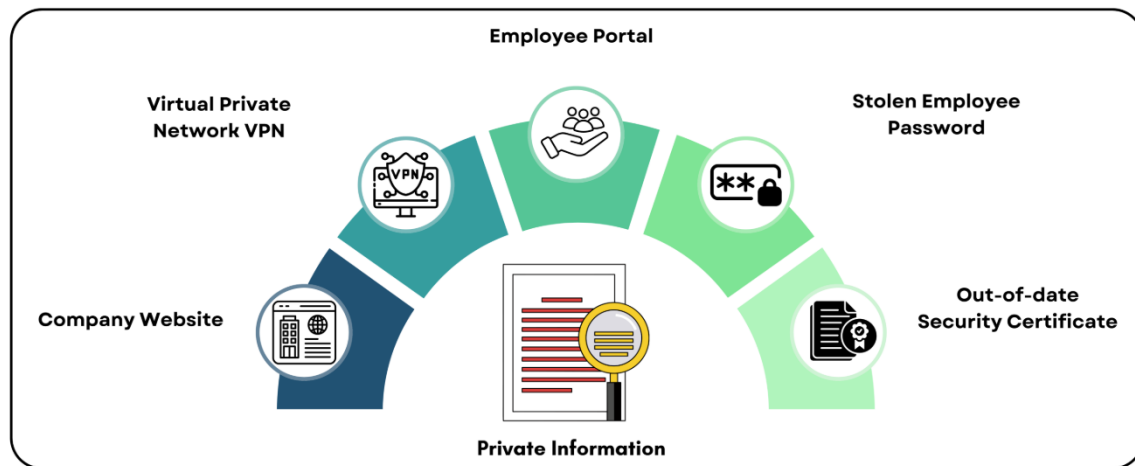


Figure 5. Components of a system attack surface, illustrating multiple exposed entry points through which private information may be accessed or compromised.

5.1 Education + IoT + wireless contexts that amplify injection and poisoning risk

LLMs and ML automation are used in education, IoT, and wireless systems to make decisions at scale, yet the mentioned settings also increase the risk of injections and poisoning due to the noisiness of inputs, their partial trustworthiness, and their tendency to be propagated into downstream operations. Yang et al. [35] demonstrate the application of LLM prompting in the automation of curriculum modeling to produce concepts and prerequisite relations, extract concepts based on a candidate list, and derive the concepts based on the candidates in a curriculum requirement using a list of concepts. Their evaluation on the MOOCCube dataset, containing 683 courses, 25,161 concepts, and 1,027 prerequisite links, shows both the benefit and fragility of prompt-driven pipelines: automatic scores vary sharply across models and prompt configurations, and relation identification errors are dominated by failures to infer implicit relations (80.23%) versus direction confusion (19.77%). This is important to security, since educational processes are very user-facing, and timely sensitivity and consecutive consumption of generated product can cause a condition in which latent instructions or malware can be carried through process steps and corrupt subsequent grading, recommendation, or curriculum networks.

Based on this risk of the workflow, interrelated IoT and wireless technologies expand the attack surface even further since natural-language or metadata fields can indirectly be used as inputs to affect system behaviour. Noman et al. [36] survey and illustrate that families of classic code-injection are extended to wireless-based IoT, in which broadcast and public accessibility facilitate easier manipulation than in wired environments. The research highlights real-world impact signals such as IoT's projected economic value of \$11.1 trillion per year by 2025, the Mirai-era outage costs including \$110 million in revenue loss for Dyn, and device recalls at scale such as 465,000 cardiac devices and 1.4 million vehicles, then provides a practical framework using a vulnerable Wi-Fi scanner on a Raspberry Pi 4 plus firmware reverse engineering and IMECA-based criticality analysis. Salmi et al. [37] expand the discussion of poisoning to the field of 5G/6G edge intelligence, showing how machine-learning-based task classification to offload communication and computing can be poisoned, and that partial recovery can be achieved with the help of anomaly detection methods. Using a 1,000-sample dataset built from ETSI 5G QoS profiles, their hybrid k-means + k-NN classifier reduces unaccepted tasks from 693 (round robin) to 14, but poisoning attacks add 278 s and 241 s to serving time; mitigation reduces damage but introduces trade-offs such as 33 false positives and 78 false negatives in one defense setting. Collectively, these works confirm that in the area of education, Internet of Things, and wireless edge system, immediate sensitivity, and untrusted data channel make injection and poisoning, not only possible but operationally significant, since the compromised outputs can penetrate automated recommendations, change the work of devices, and affect resource-allocation processes.

5.2 Reliability, geo/disaster, and digital-twin deployments (high exposure to untrusted inputs)

The need to deploy such autonomous control, geospatial analytics, and disaster response increasingly depends on retrieval-augmented generation (RAG)-empowered and agentic systems; nevertheless, such environments are very vulnerable to untrusted and rapidly evolving inputs. Zhang et al. [38] highlight this threat of the reinforcement learning in robotics and autonomous systems, showing that reinforcement reward specifications created or aided by large language models (LLMs) are fragile and manipulable when treated in the form of artefacts. Their MIRA model suggests a closed-loop system where an inner loop can optimize policy and shaped rewards by jointly optimizing policy and shaped rewards, whereas an outer metacognitive loop can keep track of the behaviour of learning and activate an LLM to revise reward factors semantically when it detects an anomaly. Empirical experiments on MuJoCo benchmark task HalfCheetah and Humanoid show that the updated rewards at run time outperform asymptotic performance and sample efficiency over

fixed/oracle rewards by encouraging continued correction of misaligned goals that could otherwise be abused or drift under noisy input. In a bid to overcome the problem of complex, real-world datasets, Peng et al. [39] consider the concept of geospatial databases where natural-language queries are converted to SQL with large GIS datasets. The Retrieval Augmented Generation (RAG) and multi-agent system built in by the investigators help in making it more robust as it retrieves metadata stored in a vector database and breaks down queries to multiple specialized agents before the creation of SQL. In urban planning database models with fifty tables and over five hundred eighty fields, the technique achieves both table-name identification accuracies of over 90% and SQL query accuracy of over 80%, much better than prompt baselines only. However, the analysts point out that geodatabases can be concerned with privacy and security because the queries made by the users, metadata, and the retrieved context can be partially distrusted. As a result, retrieval and agent coordination can be considered as the points of critical decision-making, and injection or poisoning attacks can change downstream decisions.

Stretching the limits of exposure, both disaster management and digital-twin systems rely on heterogeneous, crowd-sourced, and sensor-driven data, which is challenging to approach with overall validation. In a systematic review, Saengtabtim et al. [40] show that decision support, early warning and business continuity planning are among the applications of large language models and generative artificial intelligence methods, however, these tools are vulnerable to hallucinations, privacy breaches and malicious use of social or Internet-of-Things data streams. Lu et al. [41] describe in detail the digital twins of wetlands that combine the IoT sensors, remote sensing, simulation model, and cloud platforms in various levels of maturity, such as active intervention. This combination intensifies the consequences of manipulated inputs on automatic managerial activities. Hanny et al. [42] also demonstrate this susceptibility of disaster response by integrating textual information with the spatiotemporal aspect to classify social media relevance. The effect of multimodal model is the improvement of the macro F1 score of about 0.78 with the use of text alone to more than 0.81 with the inclusion of spatial and temporal context. Together with the above analysis, these results indicate that retrieval-augmented generation and agent pipelines need to work with large amounts of untrusted, multimodal inputs in reliability, geo-information, and digital-twin deployments, making the requirement to be robust to injection, poisoning, and misalignment one of their key necessities, not an added protection.

5.3 Healthcare, operations, and professional workflows (clinical, network ops, legal, specialized care)

Healthcare and professional workflows adopt RAG and tool-using LLM systems to reduce documentation load, improve evidence access, and support specialized decision-making, but these deployments are highly exposed to privacy leakage, prompt sensitivity, and grounding failures. In digestive disease support, Giuffrè et al. [43] argue that domain performance improves mainly through two levers, RAG over clinical guidelines and supervised fine-tuning with RLHF, yet reported baseline accuracy in gastroenterology tasks varies widely, from 6.4%–45.4% (ChatGPT-3.5) up to 40%–91.4% (ChatGPT-4), showing why strong grounding and standard evaluation are necessary. In evidence synthesis, Simmons et al. [45] operationalize this idea with a bespoke LLaMa 3-70B RAG pipeline for public health review extraction, but accuracy remains uneven across fields, with 42% of 173 extracted data fields rated accurate while 68% were rated acceptable, emphasizing that professional adoption often depends on usability plus human quality assurance rather than raw correctness.

Building on clinical workflows that require high semantic fidelity, Altalla et al. [47] show that radiology impression generation is strongly shaped by prompt design and benefits from retrieval grounding, reporting RAG outputs with BERTScore 0.92 ± 0.02 and Instructor similarity 0.90 ± 0.02 , while also noting that token limits and chunking choices drive what evidence is actually available to the model. Yip et al. [49] extend the healthcare view to integrative medicine, where systems must reconcile heterogeneous knowledge sources and languages; they propose a three-stage development strategy that starts with evidence

summarization using RAG, progresses to preference-aligned hypothetical evidence generation, and ultimately targets multimodal precision diagnosis. Their review highlights how fast medical knowledge changes, citing a shift in doubling time from 50 years (1950) to 3.5 years (2010) and a projection of 73 days (2020), which reinforces the need for retrieval-based updating and governance in clinical agents.

The same reliability pressure appears in operations and legal practice, where outputs can trigger operational actions or professional liability. Xu et al. [44] address table content-aware text-to-SQL with TCSR-SQL, combining self-retrieval via fuzz testing and RAG-style knowledge alignment, achieving 77.8% execution accuracy and 53.5% exact match on a 2,115-pair benchmark, but also showing that removing key modules can drop execution accuracy by up to 21.3%, which signals how fragile tool-grounded pipelines can be without robust retrieval and revision. In network operations, Hong et al. [46] survey LLM-based management across design, configuration, fault handling, security, and orchestration, reporting that operational settings can reach high task success or detection accuracy (for example, 91% anomaly detection in one wireless setting and configuration translation accuracies above 90% in others), yet they emphasize that hallucinations, privacy exposure from logs/configs, and latency (for example, a 13B model needing about 26 GB device memory) make validation and access control mandatory. Finally, Magesh et al. [48] empirically evaluate RAG-based legal research tools and show that hallucination is reduced but not eliminated, with Lexis+ AI at 65% accuracy and 17%-33% hallucination, Westlaw at 42% accuracy with higher hallucination, and GPT-4 producing hallucinations in 43% of responses, which directly motivates groundedness checks and citation verification in professional RAG systems.

6. MONITORING SIGNALS AND INDIRECT PROMPT-INJECTION DEFENSES

The current section critically assesses monitoring indicators and defensive strategies aimed at indirect prompt injection which is a form of attack in which its adversarial instructions have been surreptitiously interwoven into purportedly innocent inputs or recovered contextual data. These latent manipulations have the power to compromise Large Language Models (LLMs) without any direct manipulation, thus reiterating the need of sound detection and mitigation systems. The systematic identification of abnormal trend and implementation of multi-layered countermeasures to exclude unauthorized agent output manipulation are the focus of the analysis.

Modern multi-step attack detection requires the gradual importance of monitoring indicators and indirect prompt-injection defenses to guarantee system integrity involving large language models (LLM). Shaukat et al. [50] highlight the significance of event-log correlation (ELC) methods to detect multi-step attacks, where correlated events logs that are generated by various system elements are used to recreate attack chains. The mining-based methods, including clustering, anomaly detection and classification methods, such as LLMs, have promising outcomes; however, the research notes that existing tools often have a high false-positive rate, cannot identify long-range attack relationships, and can be evaded. To address these constraints, the researchers introduce hybrid LLM-assisted models that can be deployed in order to use narrative generation and dataset synthesis to augment event-log processing and workflows of the analysts. In addition, the article shows a gap in the attack-success rates benchmarks and advises the use of causal inference to future ELC methods to increase the predictive accuracy and flexibility.

Continuing this consideration, Mia et al. [51] expand the horizons of semantic relatedness prediction by creating a hybrid methodology that employs XLM -ROBERTA and convolved neural networks infrastructure to estimate similarity on sentence levels. The suggested model maintains a higher level of Pearson and Spearman correlation coefficient as compared to traditional methods because it incorporates both string-based similarity and semantic similarity metrics. The implications of the study, in spite of the fact that it focuses mainly on natural language processing, have direct implications in the defense against immediate manipulation, since the severity of semantic evaluation proves inevitable in the identification

of malicious injections in text-based prompts. The mentioned techniques can be easily scaled to larger language model-based systems by increasing their ability to detect minor input deviations, which will augment anti-injection defenses.

Based on the developed paradigm of protection against the indirect prompt injection (IPI), Bhagwatkar et al. [52] introduce a hierarchical firewall-based defense scheme that comprises of a Tool-Input Firewall, also known as the Minimizer, and a Tool-Output Firewall, also known as the Sanitizer, whose purpose is to intercept and reduce IPI incursions within agents that utilize external computational tools. This model effectively reduces attack success rates (ASR) across several benchmarks, such as AgentDojo and InjecAgent, with results showing a 0.00% ASR for the combined firewall on Tau-Bench. The suggested firewalls work by screening the bad input and disinfecting the result before it gets to the language model, significantly reducing the risk of indirect prompt injection but does not require retraining the large language model. Even though effective, the research also acknowledges the fact that more advanced attack vectors can bypass all the mentioned protective mechanisms, thus providing the rationale why defensive strategies should be constantly updated and modified. All these studies highlight the growing number of investigations that necessitate effective monitoring the systems and context-based defenses against indirect prompt injection. Increasing the complexity of systems, especially in the operational environment and in sensitive applications like healthcare and legal services, holistic defenses (correlating events, part-whole firewalls, semantic analysis, etc.) will play a fundamental role in ensuring the security and reliability of large language models-based systems.

7. ADVERSARIAL EVALUATION AND BENCHMARKS FOR AGENTS AND TOOLS

This section discusses why adversarial evaluation and benchmarking are important to Retrieval-Augmented Generation (RAG)-powered agents and tools that are prone to manipulation through carefully engineered inputs. It examines the challenges of coming up with strong benchmarks which are a true reflection of the real-world adversarial threats, and how important continuous testing is in determining the effectiveness of the defense mechanisms as the adversarial strategies develop.

7.1 Adversarial agent evaluation frameworks (multi-agent, web agents, penetration testing settings)

The shift to multi-agent large language model (LLM) systems has created various kinds of complicated security weaknesses that are regularly ignored in existing benchmark models. Kavathekar et al. [53] have filled this gap, by coming up with TAMAS, which is a benchmark specifically set up to measure emergent risk in a multi-agent environment, such as agent collusion and contradiction. Their experimental findings reveal that multi-agent systems are highly vulnerable to manipulation compared to single-model systems; impersonation attacks had a success rate of 82%, and Direct Prompt Injections (DPIs) worked successfully in 81% of the cases. Expanding the range of concepts based on the need of advanced multi-agent coordination, David et al. [54] present MAPTA, which is an autonomous penetration testing framework. Contrary to passive benchmarks, MAPTA uses a dedicated architecture, which consists of coordinator, sandbox, and validation agents to perform end-to-end exploit validation. The success rate of this methodology was 76.9% in detecting web vulnerabilities and hence it has proved that the multi-agent orchestration can successfully overcome the semantic gap between vulnerability detection and practical exploitation. Although these models are focused on the interactions at the software and logic levels, the adversarial risks also include physical and synchronization levels of networked systems. Ren et al. [55] introduce the range of adversarial detection in by providing the BAND model that detects non-reciprocal attacks in time-synchronization systems. The model measures system degradation by using a single quantity, the attack intensity, by placing physical bounds that force adversarial behaviors to satisfy a Markovian model. According to empirical studies, it

was found to be highly precise with the error between the theoretical and empirical determined intensities not surpassing 5% of most of the attack types. The findings highlight the need to incorporate physical-layer cognizance in the overall security designs to protect infrastructure on which agentic systems survive.

Lastly, as autonomous agents continue to be more entangled into multimodal environments, the risk is now moved to visual and user interface-based exploits. Ying et al. [56] suggest the SecureWebArena, which is a complete benchmark of Large Vision-Language Model (LVLM) web agents. The framework does not only assess user-level instructions, but also environment-level manipulations, including pop-ups that are malicious and false advertisements. Their study finds a significant weakness in modern LVLMs where pop-up attacks have success rates of 76% to 100% with different models. Through examining internal reasoning failures and the behaviour trajectories, the researchers prove that model specialization in its own right is not expected to provide security, thus requiring a multi-layered assessment procedure that would ensure that the implementation of autonomous agents in high-stakes web-based settings is secure. The discussion also shows that, despite the considerable practical advantages of multimodal and multi-agent large language models, the latter creates systemic weaknesses, many of which run between emergent inter-agent collusion to increasing vulnerability to visual UI exploits that are not effectively addressed through current safety measures. The results highlight the necessity to switch to multi-layered assessment systems, including TAMAS and SecureWebArena, thus addressing the task utility-adversarial robustness gap in autonomous systems.

7.2 Tool-using agent benchmarks + system-level security constraints and measurement

The deployment of large language models (LLMs) in tool-using systems has led to the creation of new benchmarks aimed at measuring possible security risks, in particular, data exfiltration and autonomous coordination. Alizadeh et al. [57] in a more recent study also studied the security posture of agentic systems using a special banking suite, thus demonstrating the ability of indirect prompt injection to threaten tool-calling agents to disclose sensitive personal information. The analysis has shown a subtle pattern of vulnerability: password information is often not easy to leak through the model, but when asked to provide sensitive information with the help of other personal information, it becomes susceptible. As a result, the results prove that tool-using agents are still strongly vulnerable to data flows manipulation and the success rate of successful attack attempts is around 20% at the range of tasks. Zhang et al. [58] explored this problem of autonomous tool use in more detail by applying the Tool-Co-benchmark that proposes the Agent-as-Tool paradigm when testing the effectiveness of sustained multi-agent cooperation. The benchmark satisfies that whilst advanced models show significant competencies in tool invocation some with success rates of over 93%, they have shortcomings in self-organization. Specifically, agents often do not switch off duplicate counterparts, which results in significant token overhead as well as compromises effective resource managing.

Continuing on the need to have coordinated agentic systems, researchers are now using these architectures to proactively protect system against cyberattacks and manage systems in real-time. Hmimou et al. [59] suggest a modular multi-agent framework which utilizes dedicated agents to verify emails, analyze logs, and evaluate IP ranges to improve the threat detection process based on semantic reasoning. Through the correlation of various indicators, such an architecture achieves an overall detection rate of 93.6% and false positives are cut by 41.3% compared to traditional signature-based methods. However, the implementation of such sophisticated security agents creates a latency and overhead to computations that have to be dealing with. Leonardi et al. [60] offer a technique of optimizing security provisions at the basic block level and ensuring the strict maintenance of system-wide schedulability. They prove that it is possible to maximize security levels, without violating real-time deadlines, in embedded and Internet-of-Things (IoT) applications by using a mixed-integer linear programming (MILP) formulation. This methodology allows the

performance overhead of security mechanisms which is often a constraint known in the study of previous agentic systems not to impair the stability of safety-critical system.

The cumulated results of such studies provide the necessary information about the dynamic environment of agentic security. First, the statistics indicate that the security posture of the tool-using agents is significantly dependent on the contextual parameters; agents that are resistant to direct attacks often fail at the vulnerabilities when the malicious instructions are introduced into the stream of data that is supposedly legitimate. Second, the study outlines a clear discrepancy, an autonomy gap, in which models are able to make tool calls in a successful way but do not have the necessary inferential capacity to stop or optimize such interactions to achieve resource efficiency. Lastly, the analysis shows that despite the fact that multi-agent LLM systems have a significant positive impact on the interpretability and accuracy of threat detection, their operational implementation is still subject to the attainment of a balance between the high level of semantics and the constraints of the systems, including timing and schedulability.

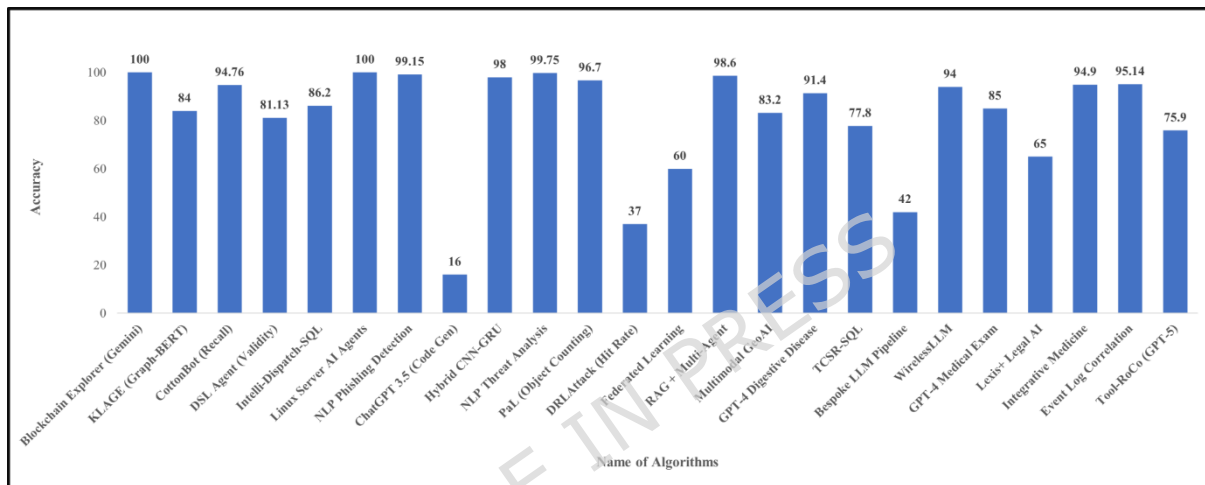


Figure 6. Performance Comparison of Algorithms Based on Accuracy Metrics.

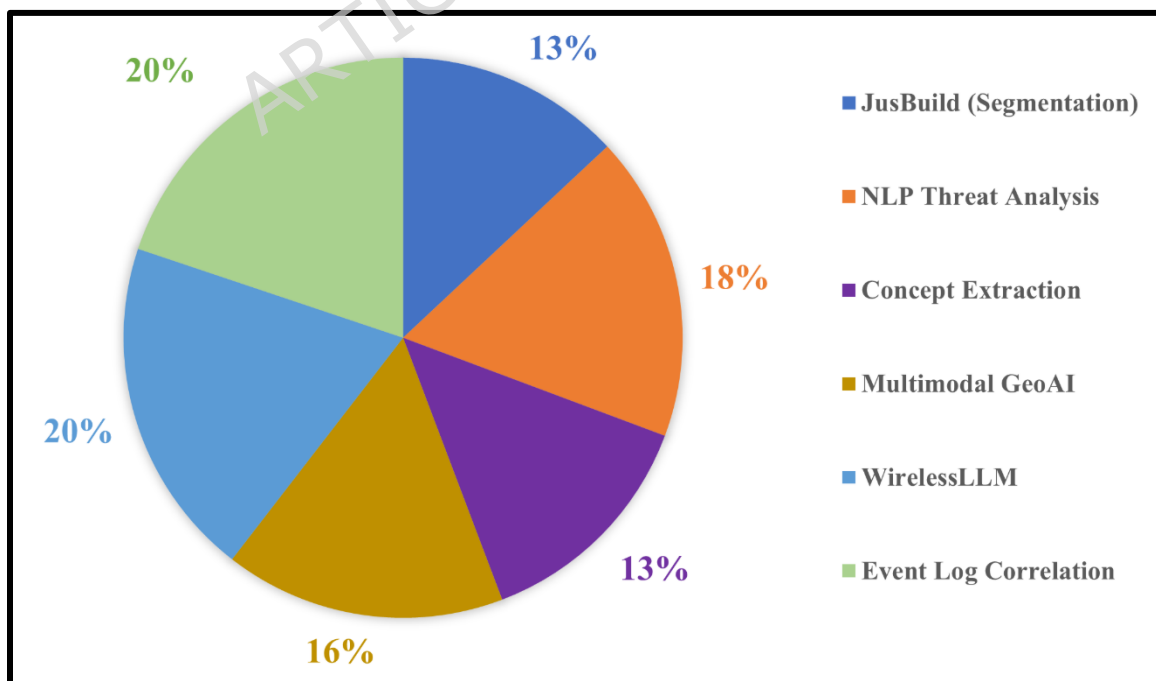


Figure 7. Comparative analysis of Precision and F1-score across different modules

In Figure 6, the accuracy performance of different algorithms is comparatively analyzed that highlights their effectiveness in different areas of application. The figure shows a diverse group of models that include models that are created in the area of blockchain exploration, natural language processing tasks, multi-agent systems, and medical applications. Results of accuracy are quite diverse; e.g., accuracy rates of some algorithms, e.g., Blockchain Explorer (Gemini) and K-LAGE (Graph-BERT) amount to almost perfect 100% whereas others, e.g., Hybrid CNN-GRU or DRLAttack, show significantly lower scores in accuracy 16% and 37%, respectively. This kind of variation underscores differences in model complexity and domain specialization and the training data, and thus the need to carefully choose an algorithm depending on the desired application scenario and performance-driven context.

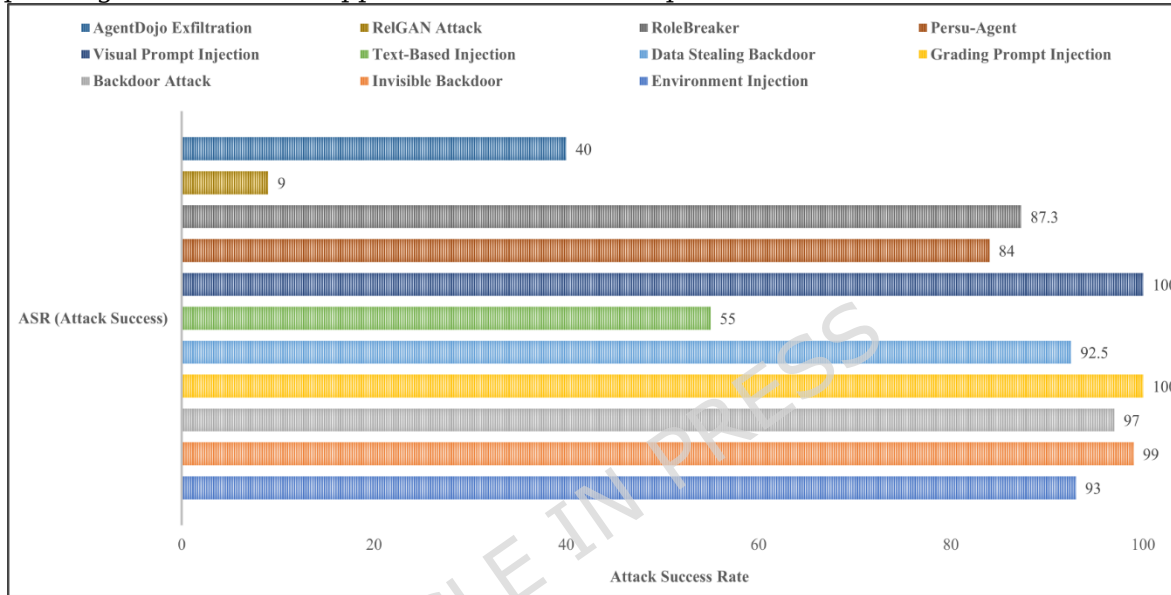


Figure 8. Attack Success Rate (ASR) analysis across diverse prompt injection, backdoor, and agent-based attack vectors targeting intelligent and multimodal AI systems.

Figure 7 represents a pie chart illustration that shows the distribution of Precision and F1-score performance of the individual system modules with percentages determined based on the normalized computed values. WirelessLLM and Event Log Correlation are the two that represent 20% of the total performance with absolute values of 98 and 99, respectively, which are indicative of better accuracy and strength. NLP Threat Analysis adds 18% of the overall performance (88), which implies a high ability to recognize and analyze textual data that is based on the threat. The multimodal GeoAI provides 16% (81) of success in the area of multimedia data fusion and geospatial intelligence. Concept Extraction and JusBuild (Segmentation) each have a 13% concept, with real performance measures of 67.48 and 65, respectively meaning moderate performance compared to the other modules. On the whole, the pie-chart represents a good visualization of a proportional contribution made by each module in terms of Precision and F1-score measures, and thus, allows intuitive comparison of system performance.

In Figure 8, the comparative analysis of the Attack Success Rate (ASR) of a variety of adversarial techniques that aim at exploiting vulnerabilities in modern AI-based systems are compared. The results demonstrate that some of the categories of attacks can have extremely high success rates, which in turn reflect serious security concerns. Visual Prompt Injection and Grading Prompt Injection reach the highest ASR of 100%, which proves their potential to control the model behaviour with the help of structured and evaluation-based prompts. Similarly, Invisible Backdoor and Environment Injection techniques exhibit apparent success, as the score in ASR values is 99% and 97%, respectively, thus indicating the discreet and insidious nature of the two attack methods. On the other hand, some attack types have

relatively lower success rates; however, they are also quite significant risks. A Text-Based Injection has an ASR of 55%, which means that it was partially resistant but still vulnerable to language-based input processing. The reports of a specific ASR in AgentDojo Exfiltration state 40%, whereas RelGAN Attack is the least expressed with 9%, which presupposes a higher resistance to the specified methods. Altogether, the figure highlights how, despite partial mitigation of certain attack vectors, the number of prompt injection and backdoor-based attacks is extremely high, and thus the need to have a well-developed defense mechanism, continuous monitoring, and flexible security systems under AI systems.

8. DEFENSIVE TECHNIQUES ACROSS CYBER & CYBER-PHYSICAL SYSTEMS

The section explores the defensive mechanisms deployed in both cyber and cyber-physical systems in the effort to reduce the adversarial threats and ensure the reliability of the systems. It discusses how these systems, including both traditional IT environments and more vital systems such as IoT networks and industrial control systems, need special-purpose defense mechanisms to mitigate particular weaknesses. It focuses on the incorporation of multilayered defense mechanisms that can accommodate both digital and physical attack vectors thus providing an all-inclusive protection in the high-risk environments.

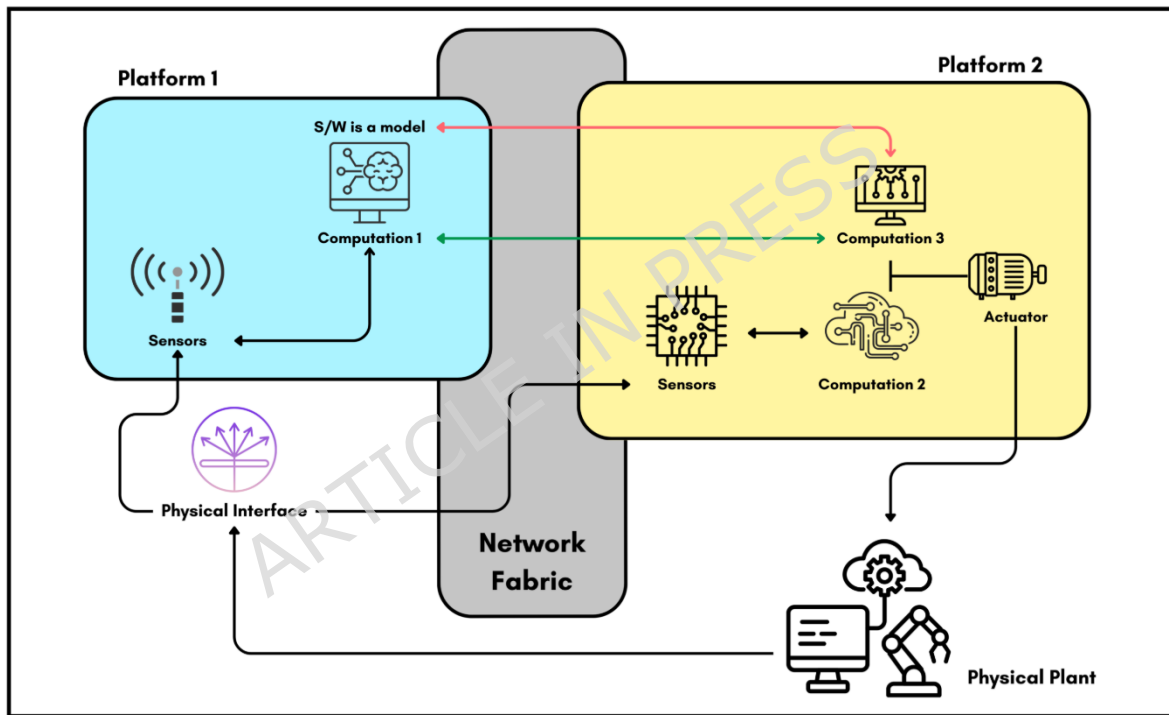


Figure 9. Cyber-physical system showing interaction between two platforms via a network fabric, with sensors, computational models, and actuators controlling and monitoring a physical plant.

The design of a cyber-physical system (CPS) is shown in Figure 9 where two platforms are connected through a network fabric. The Platform 1 will be charged with the responsibility of obtaining sensor data, and manipulating it with a computational model which may be a control or prediction module. The output of the Platform 1 is then passed through the network fabric to Platform 2. Platform 2 uses the input data to actuate actuators, which interact with the physical plant, and hence provide real-time control and feedback. The diagram also shows the data flow between sensors, computational entities and actuators thus showing how physical processes are integrated with computational models. This type of arrangement is typical of Internet-of-Things (IoT)-based systems, where continuous

communication between physical elements and digital computation is which makes the best system performance and automation possible.

Table 2. Summary of proposed models and frameworks for adversarial agent evaluation, including key algorithms, techniques, and performance metrics across various domains (IoT, healthcare, web security, etc.).

Ref No.	Proposed Model / Framework	Key Algorithms / Techniques	Key Insights & Performance Metrics
Mittal et al. [62]	AI-Augmented Framework (CI/CD to Runtime)	LSTM-based threat detection engine, Autoencoders for behavior anomaly detection, Dynamic Policy Engine for K8s enforcement	95% attack detection rate, <2s detection latency, 97% success on supply chain attacks, Automated K8s network isolation
Taher et al. [63]; Jaffal et al. [61]	TSA-LSTMARN (IoT-specific IDS)	Tunicate Swarm Algorithm (TSA) for hyperparameter optimization, LSTM-RNN for sequential traffic analysis	92.67% accuracy; 98.73% precision, Swarm optimization effectively handles IoT resource constraints and vanishing gradients
Gupta et al. [73]; Emekci et al. [67]	Holistic Impact Analysis (Survey & Taxonomy)	Taxonomy Mapping: Linking attack surfaces (e.g., poisoning) to defenses, Dual-Use Analysis: Evaluating GenAI for both SOC automation and malware generation	Defensive Utility: Automates threat reporting and secure coding, High risk of hallucinations and "jailbroken" malware creation, Lack of explainability in current defense models
Tsmindashvili et al. [72]; Shi et al. [65]	ChatTEDU (Dual-LLM), Meticulous Thought Defender	Dual-LLM: Decouples moderation (LLM-1) from generation (LLM-2), Fine-grained CoT: Dissects reasoning steps, Backward Tracing: Identifies logic anomalies	ChatTEDU: 100% attack blocking, 0.28% false positives, only 18% latency add; 93.20% accuracy (14B model), 90.13% F1, Outperforms intrinsic LLM detection by ~6%
Mo et al. [64]; Baek et al. [68]	Broken Bags Analysis (RAG Vulnerability), Hybrid CTI Classifiers (Text to MITRE)	Hybrid Poisoning: Injects minimal toxic text into knowledge bases, Sliding Window ModernBERT, Hybrid Embedding: Concatenates TF-IDF + BERT + BERTopic vectors	94% Attack Success Rate (ASR), Insight: Perplexity-based defenses fail against minimal-text injection, Model 4 (ModernBERT + TF-IDF): +10% precision improvement
Hu et al. [69]	PLLMKI (Knowledge Injection)	LLM-to-LLM Injection: LLaMA-1 generates context/knowledge; LLaMA-2 infers answer, In-context Example Retrieval	54.1% accuracy on OK-VQA, Injecting LLM-generated knowledge is more effective/cheaper than external KG retrieval for visual reasoning
Yang et al. [74]	ShellBox Framework (Interactive Honeypot)	Dynamic Error Simulation: Fakes realistic system errors, History Pruning (LRU)	81.63% success blocking prompt injections, +34.5% Turn-Level Coherence, Solves the "memory loss" issue in long-duration honeypot engagements
Shaikh et al. [71]; Kim et al. [66]	Robotic MiTM Analysis (Gray-box Threat), Multimodal Fault Detection	Traffic Interception: ARP spoofing on robot-cloud link, Early Fusion: Merges tabular (sensor) and time-series data	100% Attack Success: Induced unsafe physical actions (ignoring obstacles), 0.9900 Accuracy, 0.7500 F1 Score, Early Fusion outperforms Late Fusion for protecting physical manufacturing integrity
Chang et al. [70]	Visual Prompt-Based Mobile Learning System	Equation-Coloring, Variable-Coloring, Coefficient-Coloring Units, Pre-test and Post-test Assessment	Significant improvements in algebraic problem-solving (p-value < 0.01), 10 units higher than traditional teaching, Gamification boosts self-efficacy in students with learning disabilities

Table 2 provides a detailed description of different models and frameworks that can be used to assess adversarial agents, including the main algorithms and methods that are used in each system. It consists of LSTM-based threat detection, hybrid poisoning schemes, and multi-agent coordination of penetration testing, as well as the performance measurements,

namely, accuracy, attack success rates, and reduced false-positive. The table also demonstrates the scope of tasks included by these frameworks and includes an IoT-specific intrusion detection system and AI-based attacks on web security. Such results highlight the continued development of strong defensive strategies of multi-agent and tool-using large language model systems that act in high-risk environments.

9. RETRIEVAL ENGINEERING AND KNOWLEDGE GRAPHS AS SECURITY ENABLERS FOR RAG

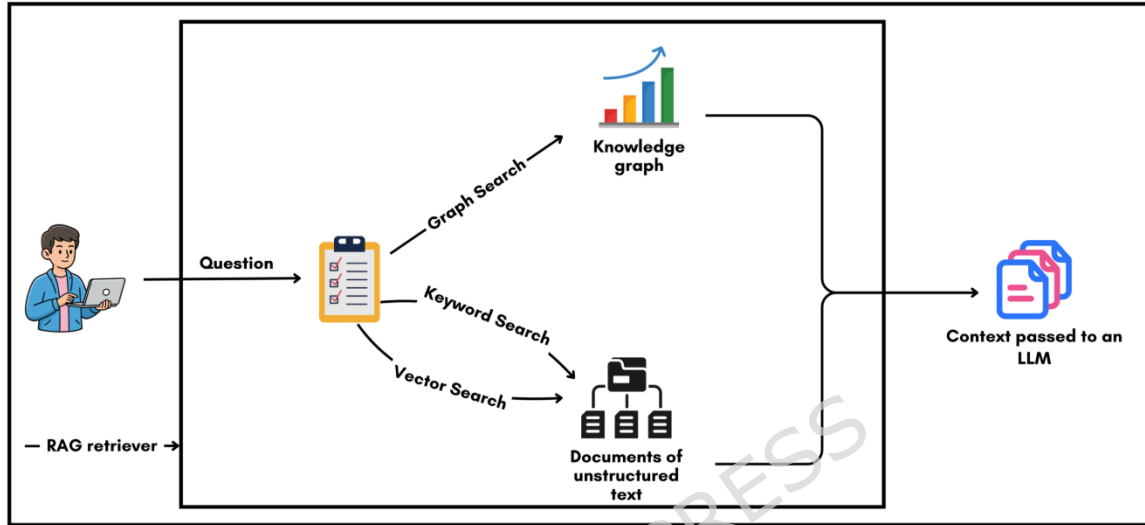


Figure 10. RAG workflow integrating Knowledge Graphs

It is shown in Figure 10 that a query is processed by a range of retrieval methods such as the graph search, key-word search, and the vector search methods. Such methods query not only unstructured textual documents but also a knowledge graph to compose the relevant context, which is then provided to a large language model that is used to generate a response. The figure explains that retrieval engineering and knowledge graphs are vital facilitators of security in Retrieval-Augmented Generation (RAG)-based systems. The systems can increase the accuracy of the retrieved context and the safety of the generated output by structuring the retrieval process around a variety of techniques and using knowledge graphs. The pool of several retrieval strategies guarantees the large language model the most significant and certified information and reduces the chances of adversarial manipulation through prompt injection or data poison.

9.1 Graph-based retrieval and hashing foundations for robust retrieval

To overcome the shortcomings of Large Language Models (LLMs) in the face of a specialized task, especially with regard to the so-called hallucinations in the context of geospatial modeling, Liang et al. built GeoGraphRAG, a graph-based Retrieval-Augmented Generation (RAG) framework. In contrast with traditional text-based RAG, this approach involved a Semantic-Geospatial Modeling Knowledge Graph (S-GMKG) which relates functional steps and data items. The system is highly structurally consistent with the help of the graph retriever which retrieves dense subgraphs based on user requirements and the solution planner which compiles the retrieved subgraphs into executable code. The proposed method was shown to perform better when it comes to automation of complex tasks to which it reached an F1-score of 0.708 with code generation that is far superior to the standard Chain-of-Thought prompting.

Swapping between generative work to effective large-scale retrieval, Xudong et al. [76] tackled the degradation of topological information that is natural to traditional deep hashing methods by proposing the Graph Hashing Network (GHash). As compared to antecedent

models which heavily relied upon constrained pairwise or triplet losses, GHash forms an alternating update protocol on graph nodes and edges thus, encompassing positional information within a large receptive field. The methodology provides very discriminative image descriptions by placing the neighborhood information inside the hash-codes. The robustness of this topology-aware approach was validated on the CIFAR-10 dataset, where it achieved a mean Average Precision (mAP) of 0.9733 for 64-bit hash codes, proving the value of structural analysis in data compression and retrieval. To increase the scale of graph-based retrieval to the multimodal contexts, Qin et al. [77] presented the Heterogeneous Graph Fusion Network (HGfN), a cross-modal image-text retrieval model. The suggested framework generates separate visual and textual graphs to represent salient regions and words, and then integrates them to create a hybrid form of representation that takes into account the asymmetry of the visual and linguistic modes. One of the key new technologies of this strategy is the so-called Dynamic Top-K Negative strategy, which chooses negative examples dynamically during training in order to optimize alignment. This approach is suitable in eliminating senseless alignments in fuzzy retrieval systems, with a Recall@1 score of 80.2% of image-to-text retrieval in the Flickr30K dataset, which further highlights the effectiveness of joint graph modeling in grounding various modalities.

9.2 KG-driven reasoning for diagnosis, mitigation, and QA workflows

Integration Knowledge graph Integration has advanced to address the shortcomings in symbolic mappings in diagnostic processes, and is being implemented towards more dynamic, semantically enhanced systems. Li et al. [78] introduced a conceptualization of electronic medical records, InKrat, as a graph over time. Through the use of anchor nodes to bridge the gap between patient encounters and the use of large language models to perform a cross-modal semantic retrieval, the researchers achieved an average accuracy improvement of about 12% on MIMIC-III datasets. Remarkably, such an approach includes a prompt-based system to form clinically interpretable rationales thus filling the gap between algorithmic forecasting and comprehension in complex diagnostic paths.

Recent academic studies have faced the problem of data sparsity in systems of decision support in the quest to broaden the use of Knowledge Graphs (KGs) not just in diagnostic settings, but also in service-mitigation settings. Similarly, Wu et al. [79] proposed KGDR-EGs to e-government services; the framework incorporates entity neighborhoods and path contexts through attention based bi-directional Long Short-Term Memory networks. This method effectively counters the cold-start problem, demonstrating a Hit Rate (HR@5) of 0.1779 compared to significantly lower baselines in sparse data scenarios. Continuing on the above semantic-embedding paradigm, Yang et al. [80] utilized deep-learning method in the field of hydropower safety management. They have come up with a multidimensional ontology of Construction Safety Hazards (CSHs) and used BERT-attention BiLSTM-CRF architecture to extract entities, achieving an F1 score of more than 95%. This system, based on sentence-BERT (SBERT) is used to directly suggest safety-mitigation interventions and thus transform experience-driven procedural workflows into data-driven governance frameworks. It is important to have powerful retrieval mechanisms to guarantee that such complex knowledge bases can be available via natural-language interfaces. Such advanced knowledge bases are not possible to maintain such powerful retrieval mechanisms which are necessary to maintain the accessibility of those knowledge bases using a natural language. Zhu et al. [81] focused on the optimization of Question-Answering (QA) processes in the ocean engineering field by alleviating the subgraph redundancy. They combine relationship consolidation with entity sequencing, and are dynamically expanded with a heuristic Beam-Search algorithm. As a result, the size of subgraphs was decreased by approximately three times without affecting the answer coverage to achieve an average of F1 of 79.10% and demonstrate that the ambiguity inherent in domain-specific queries can be addressed in a subtle way.

9.3 Entity Alignment and KG Completion for Consistency and Resilience

To make knowledge graphs (KGs) robust in terms of entity alignment and completion, there is a need to have methods that can resolve the lack of data, structural ambiguity, and multimodal complexity. In order to address the inherent limitations of single-KG methodologies, Gao et al. [82] proposed the MIKG model that maximizes the mutual information between various, disparate KGs to strengthen entity alignment. Using attribute and relational encoders under a contrastive InfoNCE task, MIKG significantly improves alignment accuracy, and shows significant resilience when attribute triples are dropped (50%). To solve the need to provide structure to complex datasets, Sun et al. [83] introduced the MG-KG model of unsupervised video anomaly detection. This method combines a Motion Guidance Network and triplet-based knowledge graph to explain video anomalies to achieve an AUC of 99.3% on the UCSD Ped2 benchmark, thus justifying the value of KG-guided interpretability to dynamic visual scenes.

Pietrzak et al. [84] evaluated CTI generated by TinyLlama and solved the problem of the quality and compatibility of high-quality synthetic intelligence with threat detection. Their hybrid detection system obtained 98.5% accuracy in differentiating CTI with human origin and AI origin and thus explains the need to strictly validate their solution within the automated intelligence pipelines. To increase the stability of generative models in long -tail scenarios, Huang et al. [85] proposed the LTGen benchmark, a template-free end-to-end pipeline that can be used to test the performance of LLMs in generating uncommon entities. They found that the use of concise KG triples in the case of hallucinations reduction is more effective using prompts than using unstructured text, which establishes a new standard in knowledge-intensive generation tasks.

The latest developments have also coincidentally focused on dedicated embedding spaces and hybrid retrieval structures to increase the accuracy of alignment. Jia et al. [86] proposed QUARTER, a quaternion graph attention network, which is enriched with LLM, and targeting temporal knowledge graphs. This model, when extending entities to a hypercomplex quaternion space and including semantic features of LLM, always beats Euclidean-based models in tasks of temporal alignment. Simultaneously, Gao et al. [87] introduced RerankKGC, which is a cooperative retrieve-and-rerank system that completes multimodal knowledge graphs. By combining embedding-based retrieval with a CLIP-based fine-tuning re-ranker, RerankKGC effectively mitigates ambiguity and underutilization of visual data, achieving up to 7.69% improvement in Hits@1 on standard benchmarks like FB15K-237. These studies collectively highlight a shift towards hybrid, semantically enriched frameworks that ensure consistency and resilience across diverse KG applications.

9.4 Embeddings, validation, and adversarial robustness in graph representations

The recent advances in the area of embedding technologies are transforming the way information is being represented and verified, especially in graphical models where efficiency and strength are of paramount concern. As a reaction to the computational complexity of dense graph representations in information retrieval, Kalogeropoulos et al. [88] introduced the Graphical Set-Based (GSB) model. The analyst uses spectral clustering and the original graph-sparsification method to recalibrate the relationships among terms to reduce the size of the graphs without losing the retrieval effectiveness. Mean Average Precision (MAP) of the method is equal to 0.2432 thus showing that spectral embeddings can remove irrelevant links, maintaining semantic coherence. Also increasing the task of retrieval efficiency to checking the knowledge already stored, Adam et al. [89] presented a traceable, LLM-based verification of RDF statements. The proposed framework uses no opaque or pre-trained models and instead, it inhibits internal knowledge in the LLM and solely uses external grounding documents; this has a precision of 88% in the evaluation of biomedical statements. The results highlight the importance of a fundamental shift in transparent and auditable validation processes in knowledge graph maintenance.

Simultaneously, the strength of these representations is also being challenged by adversarial threats in social media as well as cross-modal retrieval. Lughbi et al. [90] designed an NLP-based dashboard to classify real-time cyberattacks, where a logistic regression model produced the highest F1-score of 87.6%. The system is a visualization of the global threat

information of the social platform, which is a situational awareness tool and filters the noise of intelligence with high risk. Based on the necessity of a strong cross-modal matching, Cheng et al. [91] proposed the Adversarial Pre-optimized Graph Representation Learning (AP-GRL) model. The heterogeneity gap between image and text data is closed with the help of AP-GRL using an adaptive pre-optimization mechanism and adversarial learning with Wasserstein distance. This method obtained a MAP of 73.2% on the PKU XMediaNet dataset demonstrating that adversarial optimization may be used to impose domain-invariance representations even in multimedia contexts that are complex.

9.5 KG construction pipelines, multilingual IR, and cyber-domain analytics

Knowledge Graphs (KGs) construction has become a key pipeline to the improvement of safety and predictive performance in highly complex and high-stakes physical space. Yin et al. [92] explore the long-tail problem in self-driving as they suggest a top-down method of building KGs that reflect co-occurrence relationships between entities of traffic. Their system is capable of predicting the occlusion of objects in the graph through encoding causal and synchronous constraints, and enhancing the safety margins in the braking process within a simulation environment, thus predicting the appearance of occluded objects (like children behind barriers). Repurposing this safety-critical use case to civil engineering, Zhang et al. [93] came up with an AI-based hydraulic structure management framework. They presented the Partition Fusion Kolmogorov-Arnold Network (PFKAN) to simultaneous multi-entity relation extraction, with a score of F1 of 91.27%, the system uses a voting strategy to interpret user intent, essentially alleviating the risks of the so-called hallucinations when using Large Language Models (LLMs) to isolate and make engineering decisions.

Considering the need to have enhanced data representations in the industrial world, Zhang et al. [94] added Multimodal Large Language Models (MLLMs) to a knowledge-graph pipeline that was designed to handle vehicle operations and maintenance (O&M). The resulting model, MllmDA-KGC, uses a dual-stream transformer architecture which works with textual and visual inputs simultaneously and thus detects visual aspects, including component geometry and inter-component connectivity, that simple text-based models cannot. These multimodal representations generated an F1 score of 93.79% relation extraction and showed significant improvement in fault-diagnosis. Stokłosa et al. [95] introduced Sequential Structural Associative Knowledge Graphs (SSAKGs) by abandoning the traditional focus on semantic entities in favor of the sequential data storage. The method of encoding data sequences, representing them as transitive tournaments on the graph topology, allows effective data retrieval even without the need to undergo traditional training processes, reaching 99.6% accuracy, and aiding the extraction of the complex biological sequences, including miRNA.

Finally, to solve the linguistic issues arising with Information Retrieval (IR), Bounhas et al. [96] presented a morpho-semantic knowledge graph, CAMS-KG, that is specific to classical Arabic. The structure reduces the ambiguity caused by the lack of diacritics in the Arabic sources with the connection of morphological analysis and semantic association. To go further than the primitive stemming, the researchers used a resilient lemma-based indexing scheme, which, as it is shown, provides a significantly better performance compared with the traditional baselines, reaching a Mean Average Precision (MAP) of 52.4% percent. Together, these studies map the shift of knowledge-graph pipelines online to movement, multimodal, and linguistically complex systems that support advanced analytics across the spectrum of fields of study.

9.6 Discussion and Research Gaps

Within the context of agentic workflow paradigms, even in the present-day state, both the major improvements have remained quite vulnerable to the indirect injection of prompts in retrieval pipelines, and the solutions have not suggested any strong, low-latency defensive strategies, which are adequate to real-time cyber-physical systems. Besides, the current Retrieval Augmented Generation (RAG) systems often do not ensure the protection of the

agent tool interface against corpus poisoning and visual exploits, thus it represents a security gap in dealing with multimodal and long tail querying of domains.

Table 3. Analysis of research gaps and actionable steps

Research Gap & Sources	Gap Description	Actionable Steps
Indirect & Multi-step Injections Sources: Kavathekar et al. [53], Alizadeh et al. [57], Yang et al. [74]	Agents miss "indirect" injections (e.g., emails, databases) or attacks across multiple steps.	Use a transformer-based model to analyze retrieved data chunks before processing. Sanitize outputs from external tools to strip hidden commands.
Corpus Poisoning Sources: Mo et al. [64], Adam et al. [89]	RAG systems vulnerable to poisoning by toxic text mimicking queries.	Mix similar and diverse results during selection to reduce impact of poisoned data. Enforce strict checks to identify and block outlier nodes in the search process.
Security Latency in Real-Time Systems Sources: Leonardi et al. [60], Shaikh et al. [71], Kalogeropoulos et al. [88]	Defense mechanisms introduce unacceptable latency in real-time systems.	Use lightweight, efficient classifiers and fast vector search methods to maintain low latency.
Lack of Robustness in Multimodal Agents Sources: Ying et al. [56], Zhang et al. [94], Gao et al. [87]	Agents are vulnerable to visual exploits (e.g., overlays, pop-ups).	Ensure that non-essential metadata is stripped, and inputs from visual elements are validated like code.
Failure on Long-Tail Queries Sources: Liang et al. [75], Huang et al. [85], Yin et al. [92]	LLMs struggle with rare entities or specialized domains.	Broaden search queries to cover rare terms and utilize efficient vector search methods to retrieve sparse, valuable domain knowledge.
Unsecured Agent-Tool Interfaces Sources: David et al. [54], Zhang et al. [58]	Tool interface is often trusted, allowing malicious outputs to hijack control.	Implement two-tier firewalls: one to secure inputs to the agent and another to filter outputs from external tools.
Insufficient Verification of Synthetic Intelligence Sources: Pietrzak et al. [84], Baek et al. [68]	Detectors struggle to distinguish AI-generated threats from real ones.	Train classifiers to recognize specific patterns of synthetic content and segregate potential AI-generated threats.

10. PROPOSED ARCHITECTURE: AEGISRAG-RADMS

The review presents a theoretical framework named AegisRAG-RADMS to counter the grave vulnerabilities of the current multi-agent systems and insufficient robust defenses in retrieval-augmented processes. The idea behind the architecture is that a graph-based reasoning and multi-agent coordination can be combined to generate a powerful security layer, which is specially designed with the objective of detecting and mitigating advanced adversarial attacks without slowing down the system in a real-time environment.

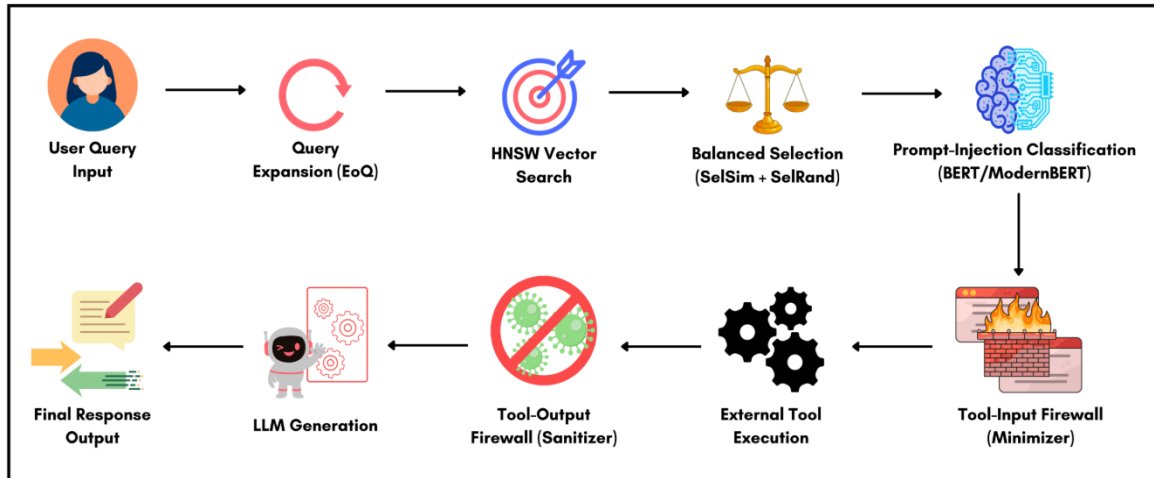


Figure 11. AegisRAG-RADMS Pipeline Overview: A comprehensive workflow for query processing, balancing efficiency, accuracy, and security through advanced algorithms, classification models, and firewall systems.

The AegisRAG-RADMS architecture is a proposed retrieval-augmented generation pipeline as presented in Figure 11, which has been developed as a conceptual way of curbing the threat of adversarial attacks prior to them influencing the generative model. This begins with the user query entry that is processed by the Query Expansion (EoQ) to expand the query semantics and prevent potential evasion strategies. Subsequently, a contextual information is retrieved with the assistance of an HNSW vector search in a balanced selection regime. This step, using high-similarity (SelSim) and randomized (SelRand) selection, is supposed to decrease the likelihood of corpus poisoning, and also to preempt the system converging on groups of manipulated data. The second important pre-processing defense is the implemented Prompt-Injection Classification that applies BERT or ModernBERT to analyze the retrieved context to detect adversarial artifacts and is developed to ensure that malicious prompts are prevented in the first section of the pipeline.

Algorithmic Mechanics of Pre-Processing Defenses: To expound on the technical implementation of the early pipeline, the Query Expansion (EoQ) module works by creating several semantic paraphrases of the original user query to calculate an aggregated centroid vector to be used in the HNSW search. This stochastic perturbation breaks brittle adversarial suffixes based on the exact and worst-case matching of tokens mathematically. After retrieval, the Prompt-Injection Classification is based on the bidirectional attention of BERT or ModernBERT. After subjecting the retrieved text chunks to the model's transformer encoder layers, the model produces a dense contextual embedding, which is then passed through a special classification head. Assuming the probability of malicious instruction-tuning data or jailbreak signatures is computed, when this probability is above a hard empirical limit, the bad chunk is mathematically marked and quarantined before it can be loaded into the context window of the agent.

The second phase of the architecture is devoted to the reinforcement of the agent to tool interface with the help of Dual-Firewall approach. A Tool-Input Firewall (The Minimizer) filters out unnecessary data and makes a significant amount of validation of input structures before the External Tool Execution, minimizing the attack surface. Upon completion of the operation of the tool, a Tool-Output Firewall, which is marked Sanitizer, directly captures the raw output, emptying the indirect injections and possibly malicious payloads to prevent corruption of the logic of the agent. The checked and filtered information is then forwarded to the LLM Generation module which subsequently generates the Final Response Output. The theorized nature of such a continuous verification loop is to assist in ensuring that the

system maintains high-fidelity performance, as well as being resistant to direct attacks and more complex and multi-step indirect injection attacks.

Exploit Mechanics vs. Dual-Firewalls: To see the need of this dual-firewall strategy, think about the dynamics of a multi-step exploit. In this case an attacker may initially perform an indirect prompt injection by planting a disabled, malicious code within an apparently harmless document within the vector database. Retrieved, this tainted context tries to fool the LLM into making an unauthorized tool call: e.g., trying to query a backend database with bad parameters. The Minimizer (Tool-Input Firewall) will intervene at this point; through the enforcement of pre-defined API schemas, allowlists and least-privileged parameter validation, the unauthorized request is blocked prior to the external tool being executed. An invalid access to a compromised external resource (like a poisoned third-party web API) by a legitimate tool call, on the other hand, may result in the returned data containing secondary injection commands to hijack the ultimate generation step of the agent. The Sanitizer (Tool-Output Firewall) prevents this latter vector by cleaning the raw tool output of runnable code and obscuring sensitive information prior to re-reading by the LLM. These firewalls combined break the chain of attack at the outward invocation and inward retrieval phases, eliminating multi-step attacks which are easy to get around simple input filtering.

10.1 Verification and Retrieval Primitives that Support Secure Grounding

Safe grounding in retrieval-augmented systems is based on the ability to check facts and retrieve the necessary evidence with the maximum level of accuracy, particularly in the case when the risk of hallucination exists. In an effort to address the validity issue of complex claims, Chen et al. [97] developed Concept-HGN as a heterogeneous graph network, which is capable of linking multi-granularity assertions i.e., between claims and fine-grained entities through external commonsense concepts on YAGO. This approach allows one to reason dynamically based on semantic dependencies, with an accuracy of labels of 80.26 per cent on the FEVER dataset in turn demonstrating that concept injection in a structured manner is quite useful in strengthening verifications. To supplement this need of narrow retrieval in large scale visual fields, under the presumption that it is possible to learn discriminative binary codes, Zhang et al. [98] proposed Deep Collaborative Graph Hashing (DCGH) a framework that trains discriminative binary codes through joint optimization of visual and semantic embeddings. DCGH bridges the semantic gap in image retrieval by constructing a shared latent space, which can be effectively utilized by graph convolutional networks to bridge the gap to continually reach improved results on benchmarks such as CIFAR-10.

The reorientation of generalized retrieval techniques to domain-specific design and diagnostic pipes is a sign of the significance of dynamic knowledge graphs in the solving of specialized rules in the presence of a weak supervision. Xiong et al. [99] suggested the DR-RAG retrieval-augmented generation system which is an aviation manufacturing system, that is aviation manufacturing specific and applies dynamic knowledge graphs and digital twins simulation. It is proposed that the suggested system, which relies on the rule-based reasoning and a closed-loop feedback mechanism, will reduce the number of design cycles required and allow a more significant level of transparency in the decision-making process, resulting in a Recall@10 of 75.43% on difficult queries in the form of associations. Likely, Jin et al. [100] had to deal with the unavailability of annotated data in medical diagnostics, and they proposed DGWSDH, a more dynamic scheme of deep hashing graphs that are currently used in Whole Slide Image (WSI) diagnostics. The end-to-end design constructs sparse dynamic graphs that reflect the relationship between patches and can classify and retrieve well with just slide-level labels; it significantly improves F1 scores and by up to 7.74% in multiple cancer datasets.

AegisRAG-RADMS model is a comprehensive, conceptual verification and retrieval primitives model that simplifies to a single security model suggested to solidify agentic workflows in defense of adversarial attacks. The purpose of AegisRAG-RADMS is to make sure that the contexts that are being retrieved are semantically checked prior to ingestion, by

incorporating the concept-verification principles of Concept-HGN [97]. It also employs efficient hash and dynamic graph building methods, inspired by DCGH [98] and DGWSDH [100], to facilitate its HNSW Vector Search to provide fast, scalable, and noise resistant context search. More to the point, the Balanced Selection (SelSim + SelRand) mechanism implemented into AegisRAG-RADMS serves as a suggested countermeasure to the vulnerabilities of the former, which dynamically diversifies the sources of evidence, and seeks to practically resolve the issue of corpus-poisoning in high-stakes areas, such as aviation and healthcare. This is a multilayered solution, a combination of a stringent method of prompt-injection classifications with a Dual-Firewall execution policy, constitutes a hypothetical defense perimeter to safeguard the retrieval pipeline and the gate between the agent and the tool.

10.2 Risk Reasoning, Evaluation, and Scalable Training for Graph-Supported Defenses

Combining real-time information with graph-based reasoning is transforming the stagnant analysis of risk regimes into moving and actionable intelligence, especially in areas sensitive to risk. The first to create a dynamic hazard analysis approach to construction site was Zhang et al. [101], in which real time measurements are fed through computer vision into a knowledge graph. To capture the changing risks, the system dynamically changes topology by modeling a hazard triggering ability and susceptibility, and the system is nearly perfectly accurate at label-recognition in the field. Large-scale multimedia retrieval is an expression of this shift of large-scale, graph-based evaluation where indexing is essential. Yao et al. [102] introduced Efficient Supervised Graph Embedding Hashing (ESGEH) scheme, a scheme that employs class labels to train a shared semantic subspace to be applied in the cross-media retrieval process. ESGEH breaks down the intermediate terms to do cost-effective computation of the Laplacian, and, therefore, high-fidelity retrieval at scale can be done without reducing speed, demonstrating that the algorithm complexity has been reduced by a factor of two.

These sound evaluation structures require special data sets and training procedures to support them. In order to address this information gap in medical risk analysis, Singh et al. [103] developed a special Diabetes Knowledge Graph to aid in link prediction. The curated benchmark presented demonstrated that the latest embedding models, e.g., pRotatE can reach a Mean Reciprocal Rank (MRR) of 0.74 and it exceeds the performance of baseline algorithms by a significant margin due to which they might be used as a resource of critical value in the development of prediction algorithms in healthcare. Based on the principle of distributed and secure training, Yan et al. [104] proposed FGNN-SGR, a federated learning framework that does text-image retrieval. This algorithm, using similarity-graph reasoning on local clients and a modular weighted-aggregation computational mechanism on the server, preserves data privacy and an Rsum of 528.9 on MSCOCO, demonstrating that complex graph neural networks can be trained jointly on multiple sources of decentralized origin without any security loss.

AegisRAG-RADMS combines the current developments in risk rationale and programmable evaluation with the aim of offering a dynamically adaptive and strictly validated defense architecture. According to the dynamically updated topology presented by Zhang et al. [101], the search of vectors in HNSW and the controlled selection process are developed to change the retrieval paths based on the novel threat patterns and, in this manner, to mitigate the activation of the poisonous sets of contexts. The lightweight Tool-Input Firewall (Minimizer) is founded on the principles of efficiency that are based on ESGEH [102] such that the security checks are suggested to be proportionally related to the extent of interaction and therefore low latency is actually preserved by real-time agents. In addition, the federated aggregation principles of FGNN-SGR [104] update a proposed Prompt-Injection Classification training plan, providing the classifier to internalize information of other, decentralized threat-intelligence feeds without exposing sensitive internal information. Through these scalable and graph-based primitives, the AegisRAG-RADMS system is

hypothetically capable of implementing a continuous risk evaluation process and thus safeguard agentic behaviors in dynamic adversarial environments.

Computational Overhead and Latency: Although the AegisRAG-RADMS architecture offers a solid defense perimeter, the very implication of the introduction of such mechanisms is the introduction of computational overhead which must be offset against system responsiveness. The pre-processing phase, based on BERT or ModernBERT to classify prompts by injecting them, is associated with inference latency that increases with the size of the retrieved context. Despite the speed optimization of ModernBERT, processing delays can still be experienced by real-time enterprise or robotic agents. Additionally, the Dual-Firewall system will introduce structural validation expenses; Minimizer needs to perform strict schema validation prior to the execution of the tools, and Sanitizer needs to perform pattern matching and data scrubbing after the execution. In order to make this viable in real-time, it is hypothesized that system designers adopting this architecture will have to avoid these trade-offs carefully through the use of lightweight, quantized classifier models, parallelization of the input/output validation process, and caching verified responses of tools in order to reduce repetitive processing overhead.

11. CONCLUSION

The present survey is a review of the security situation pertaining to retrieval-enhanced with generation (RAG)-enabled, tool-using large language model (LLM) agents. It concludes that convergence of retrieval, reasoning and action execution leads to a weak, broadened attack surface that is susceptible to retrieval poisoning and indirect prompt injection, and therefore it breaks the constraint of traditional model-based vulnerabilities. The scholars believe that the existing protective systems cannot be considered good enough to implement them in high stakes applications in fields like healthcare, legal practice, and the Internet of Things (IoT). The main reason of this failure is that there is no system-level contextual awareness that exposes the agents to multi-step exploits in which malicious code is concealed within the retrieved documents or within tool output. To overcome these shortcomings, the study suggests a conceptual reference architecture called AegisRAG-RADMS. This theoretical framework hypothesizes a defensive implementation that is a combination of formidable HNSW retrieval, embedding-based query expansion (EoQ) and a newly designed dual-firewall system, which comprises of an input Minimizer and an output Sanitizer, with the objective of subjecting all external agent-tool interfaces to stringent validation to be subjected to future empirical testing. In addition, the review suggests that it is necessary to go beyond the fixed benchmark evaluation of potential security structures to add on-going adversarial testing and graph-based checks. The correctness of this recommendation can be justified with the facts that knowledge graphs are useful in reducing hallucinations and semantic consistency in the dynamic operation environment. Lastly, the research verifies that provable soundness in self-directed actors involves a governance paradigm that combines real time threat detection and unaltered audit trail, which is mandatory to assist in reducing adversarial control of physical and cyberspace resources.

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